de mathématique et d'informatique
Université de Strasbourg

Internship report

The Riemann-Hilbert correspondence

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2026

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This report is the result of an internship I conducted at the Mathematical Institute at Freiburg im Breisgau, Germany, under the supervision of Prof. Dr. Annette Huber-Klawitter between May 25th and July 25th of 2026.
I wrote out a "blog" at the adress `https://grenadine-las.github.io/site-web/feuille_de_route_memoire_fribourg.html` during the internship. I will summurize its content in this section : what I have studied, in what order, the talks I gave and the ones I attented to, etc.

Before arriving in Freiurg im Breisgau, I attended to some sessions of the seminar on algebraic $\mathcal{D}$-modules co-organized by Annette Huber-Klawitter and Ben Snodgrass, see `https://home.mathematik.uni-freiburg.de/arithgeom/lehre/ss26/programm.pdf` for the program. Moreover, during a video interview, I had been given as a homework to get some knowledge on sheaf theory in order to prepare a talk in the aforementioned seminar (talk 8). It is also essential for what I was about to study : the Riemann-Hilbert (*RH* for now on) correspondence.

The firsy day I arrived, after a tour of the institute and of the (few, it was a holiday week) staff, the big picture of the mathematical content of the internship had been dropped : the Riemann-Hilbert correspondence, first on a complex manifold (IV.2), using the notion of connections and local system, and then in a more algebraic way with $\mathcal{D}$-modules (VI), using perverse sheaves and derived categories. The final goal would be to relate this setup with periods coming from number theory. This is the last part of this report, see VII.3. This link is made in particular with the Gauss-Manin connection.
Finally, was mentionned the specific case of what would be the computation for the Riemann surface $\hat{\mathbb{C}} \setminus \{s_0, \ldots, s_k\}$.

The first three week were dedicated to the (re)familiarization of the terms : first some historical background to the RH correpondence, analytic and algebraic complex geometry, category theory, sheaf theory ... a lot of this can be found in III.
In parallel, a friend of mine recommanded me the fantastic book of T. Szamuely [Sza12], in which the RH correpondence is adressed in dimension 1, and references given for further developpement.

I gave the talk "Background in sheaf theory" and was asked to do the 10th one intitled "Riemann-Hilbert correspondence" : this was the one I was given as a particular example. I had some trouble understanding the meromorphic lifting of connections, as expressed in the fundamental work of Deligne [Del70]. See [Kat76] for an introduction.

After this, it was time to switch in an algebraic setup. First, study the correpondence in terms of varieties over $\mathbb{C}$, using the GAGA machinery.
Then in the language of $\mathcal{D}$-modules, stated in the derived categories. The whole theory, with the one related to (co)homology, took me some time to precess. Everything needed to be translated in these terms, and for example, the RH correspondence as stated in [HTT08] only appears Chapter 7, knowing that some contents in every previous chapter was necessary. I still did not make through every details ; that is one reason why this section is so sketchy in the report.

Finally, the arithmetic-geometry-related part with periods. The first thing to understand was the Gauss-Manin connection, its different constructions and how it was important for the study of periods around singularities. I only understood few of these results : I had to set aside the theoretical aspect, as I have not found anyone who has written down the details (except for particular case as the Legendre family of elliptic curves). The horizon in this direction seems wide, and some of it will be discussed in the conclusion.

Aside, I had the chance to participate to some conferences : three sessions of the "algebra seminar" at the Institute, the "BNS" (for *basic notion seminar*), a colloquium, and back in Strasbourg, the nineth meeting of the "Rhine Seminar on Transcendence", and finally the 60th year anniversary of the IRMA.

III Prerequisites

III.1 Category theory

For the first three subsubsection, we follow the presentation by Assem in [Ass22] and try to summurize it for the purpuse of this paper. We add the precise reference for the aforementioned book to each paragraph and each result that would need a proof.
Of course, we expose the basic notions of categories, so many books contains such material.

III.1.1 First notions

III.1.1.a. Starting point

[[Ass22], I].

Definition III.1.

A *category* $\mathscr{C}$ is given by the data of :

1. a class $\mathrm{Ob}(\mathscr{C})$, called the *class of objects* of $\mathscr{C}$,
2. for every couple of objects (x, y) of $\mathscr{C}$, a class $\mathrm{Hom}_{\mathscr{C}}(x, y)$, called the *class of morphisms* from x to y,
3. for every triplet of objects (x, y, z) of $\mathscr{C}$, a function

$$\begin{aligned} \mathrm{comp}_{(x,y,z)} : \quad \mathrm{Hom}_{\mathscr{C}}(x, y) \times \mathrm{Hom}_{\mathscr{C}}(y, z) &\longrightarrow \mathrm{Hom}_{\mathscr{C}}(x, z) \\ (f, g) &\longmapsto g \circ f \end{aligned}$$

   satisfying the following proprerties :

   a. (*Associative law*) for every quartet of objects (w, x, y, z) of $\mathscr{C}$, if $(f, g, h) \in \mathrm{Hom}_{\mathscr{C}}(w, x) \times \mathrm{Hom}_{\mathscr{C}}(x, y) \times \mathrm{Hom}_{\mathscr{C}}(y, z)$, then $(h \circ g) \circ f = h \circ (g \circ f)$,
   b. (*Identities*) for every object x of $\mathscr{C}$, there exists an element 1_x of $\mathrm{Hom}_{\mathscr{C}}(x, x)$, so that for every couple of objects (w, y) and every morphisms $(f, g) \in \mathrm{Hom}_{\mathscr{C}}(w, x) \times \mathrm{Hom}_{\mathscr{C}}(x, y)$, then $1_x \circ f = f$ and $g \circ 1_x = g$.

   $g \circ f$ is called the *composition* of g with f.

Remark III.1.

- We will use the following notations : $x \in \mathrm{Ob}(\mathscr{C})$ if x in an object of $\mathscr{C}$, $f : x \to y$ if f is an element of $\mathrm{Hom}_{\mathscr{C}}(x, y)$.
- If $f : x \to y$, x is called the *domain* or *source* of f and y the *codomain* or *target* of f.
- For our purpose, it is sufficient to consider that for every $(x, y) \in \mathrm{Ob}(\mathscr{C})^2$, $\mathrm{Hom}_{\mathscr{C}}(x, y)$ is a set. In this case, the category is said to be *locally small.*

Example III.2.

We only give examples that will be useful in this text. For an extensive list of examples, see [Chapter 1, Examples 2.2].

1) $\mathscr{S}et$, $\mathscr{A}b$, $\mathscr{R}ings$ will respectively denote the category of sets, abelian groups, commutative

rings with unit, with clearly associated morphisms.

2) Let $(X, \mathcal{T})$ be a topological space. We define its topology category $\mathcal{T}_X$ having objects $\mathrm{Ob}(\mathcal{T}_X) = \mathcal{T}$ and morphism $\mathrm{Hom}_{\mathcal{T}_X}(U,V) = \begin{cases} U \hookrightarrow V & \text{if } U \subseteq V \\ \emptyset & \text{otherwise.} \end{cases}$

Definition III.2.

Let $\mathscr{C}$ be a category.
A category $\mathscr{D}$ is said to be a *subcategory* of $\mathscr{C}$ if the following are satisfied :

1. every object of $\mathscr{D}$ is an object of $\mathscr{C}$,
2. for every $(x,y) \in \mathrm{Ob}(\mathscr{D})^2$, then $\mathrm{Hom}_{\mathscr{D}}(x,y) \subseteq \mathrm{Hom}_{\mathscr{C}}(x,y)$,
3. for every $x \in \mathrm{Ob}(\mathscr{D})$, then $1_x \in \mathrm{Hom}_{\mathscr{D}}(x,x)$,
4. for every $(x,y,z) \in \mathrm{Ob}(\mathscr{D})^3$, every $(f,g) \in \mathrm{Hom}_{\mathscr{D}}(x,y) \times \mathrm{Hom}_{\mathscr{D}}(y,z)$, then $g \circ f \in \mathrm{Hom}_{\mathscr{D}}(x,z)$.

$\mathscr{D}$ is said to be a *full subcategory* of $\mathscr{C}$ if in addition, for every $(x,y) \in \mathrm{Ob}(\mathscr{D})^2$, then $\mathrm{Hom}_{\mathscr{D}}(x,y) = \mathrm{Hom}_{\mathscr{C}}(x,y)$.

III.1.1.b. Functors

III.1.1.c. Natural transformations

III.1.2 (Co-)limits

The theory of (co)limits is way more wide that what we present here. We again restrict to out purpuse.

III.1.2.a. (Co-)kernels

Definition III.3.

Let $\mathscr{C}$ be an additive category, $f : X \to Y$ a morphism.

1) A *kernel* of f is a couple (U,u) with $U \in \mathrm{Ob}(\mathscr{C})$ and $u : U \to X$ such that :
   a. $fu = 0$,
   b. for every couple (U',u') with $U' \in \mathrm{Ob}(\mathscr{C})$ and $u' : U' \to X$ satisfying $fu' = 0$, there exists a unique $g : U' \to U$ so that $ug = u'$:

$$\begin{tikzcd} U \arrow[r, "u"] & X \arrow[r, "f"] & Y. \\ U' \arrow[u, dashed, "\exists! g"] \arrow[ur, "u'"'] & & \end{tikzcd}$$

2) A *cokernel* of f is a couple (V,v) with $V \in \mathrm{Ob}(\mathscr{C})$ and $v : Y \to V$ such that :
   a. $vf = 0$,
   b. for every couple (V',v') with $V' \in \mathrm{Ob}(\mathscr{C})$ and $v' : V' \to X$ satisfying $v'f = 0$, there exists a unique $g : V \to V'$ so that $gv = v'$:

$$\begin{tikzcd} X \arrow[r, "f"] & Y \arrow[r, "v"] \arrow[dr, "v'"'] & V \arrow[d, dashed, "\exists! g"] \\ & & V'. \end{tikzcd}$$

III.1.2.b. Canonical factorisation

See [Ass22, VII §4].

Definition III.4.

Let $\mathscr{C}$ be an additive category and f a morphism in $\mathscr{C}$. We define :

1) the image of f as the kernel of the cokernel of f,
2) the coimage of f as the cokernel of the kernel of f.

Proposition III.1 ([Ass22], Proposition 4.3).

Let $\mathscr{C}$ be an additive category in which every morphism admits a kernel and a cokernel. Let $f : x \to y$ be a morphism.
Then, there exists a unique morphism $\bar{f} : \mathrm{Coim}(f) \to \mathrm{Im}(f)$ so that $f = j\bar{f}p$, meaning that the following diagram commutes :

$$\begin{tikzcd} \mathrm{Ker}(f) \arrow[r, "k"] & X \arrow[r, "f"] \arrow[d, two heads, "p"'] & Y \arrow[r, "c"] & \mathrm{Coker}(f) \\ & \mathrm{Coim}(f) \arrow[r, dashed, "\exists! \bar{f}"] & \mathrm{Im}(f) \arrow[u, hook, "j"'] & \end{tikzcd}$$

This decomposition is called the *canonical factorisation* of f.

III.1.3 Some special types of categories

III.1.3.a. Additive categories

See [Ass22, VI §2, VII §2].

Definition III.5.

A category $\mathscr{C}$ is said to be a *additive* if :

1. for every $x, y \in \mathrm{Ob}(\mathscr{C})$, the set $\mathrm{Hom}_{\mathscr{C}}(x, y)$ is a abelian group,
2. the composition is left and right distributive over the sum,
3. it admits finite products and coproducts.

III.1.3.b. Abelian categories

See [Ass22, VIII §2, 3].

Definition III.6.

Let $\mathscr{C}$ be a category, so that every morphism admits kernel and cokernel, and f a morphism.
f is said to be *strict* if its canonical factorisation $\bar{f}$ (see III.1) is an isomorphism.

Proposition III.2 ([Ass22], Theorem 2.8).

With the same notation as the previous definition, the morphism f is strict if, and only if there exist a kernel j and a cokernel p such that $f = jp$.

One can show that in this case, $j = \mathrm{Im}(f)$ and $p = \mathrm{Coim}(f)$ (see Corollary 2.10 in [Ass22]).

Corollary III.3 ([Ass22], Corollary 2.9).

A strict morphism f that is an epimorphism and a monomorphism is an isomorphism.

Definition III.7.

An additive category is said to be *abelian* if :

1. every morphism admits a kernel and a cokernel,
2. every morphism is strict.

III.1.4 Homology, cohomology, homotopy

The categorical approach to (co)homology have historically been motivated by algebraic topology and geometry. It is entirely normal for the notation to remind the reader of concepts already encountered in applications.
We follow [Ass22, IX§5].
In this subsubsection, $\mathscr{C}$ is an abelian category.

III.1.4.a. Complexes of (co)chains

Definition III.8.

1) A complex of chains is the data of $C_\bullet = (C_n, d_n)_{n\in\mathbb{Z}}$ with the C_n's objects of $\mathscr{C}$ and $d_n : C_n \to C_{n-1}$ such that for every $n \in \mathbb{Z}, d_n \circ d_{n+1} = 0$. We will denote :

$$C_\bullet : \dots \to C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \to \dots$$

2) A complex of cochains is the data of $C^\bullet = (C^n, d^n)_{n\in\mathbb{Z}}$ with the C^n's objects of $\mathscr{C}$ and $d^n : C^n \to C^{n+1}$ such that for every $n \in \mathbb{Z}, d^n \circ d^{n-1} = 0$. We will denote :

$$C^\bullet : \dots \to C^{n-1} \xrightarrow{d^{n-1}} C^n \xrightarrow{d^n} C^{n+1} \to \dots$$

The rest of the text will mainly use the second definition, so we will state everything in this term, but the dual can easily be found by the reader.

Definition III.9.

We say that a complex of cochains $C^\bullet$ is bounded above (resp. bounded below) if there exists $i \in \mathbb{Z}$ such that for every $j \geq i$ (resp. $j \leq i$), $C^j = 0$.
$C^\bullet$ is said to be bounded if it is bounded below and above.

Remark III.3.

The condition $d^n \circ d^{n-1} = 0$ means $\mathrm{Im}(d^{n-1}) \subseteq Ker(d^n)$ (when this notation makes sense in

$\mathscr{C}$).

Definition III.10.

Let $C^\bullet$ be a complex of cochains.
$C^\bullet$ is said to be exact if for every $n \in \mathbb{Z}$ the following holds : $\mathrm{Ker}(d^n) = \mathrm{Im}(d^{n-1})$.

Example III.4.

Let $C^\bullet$ be a complex of cochains and $n \in \mathbb{Z}$. We say that $C^\bullet$ is *concentrated in degree* n if $C_m = 0$ for $n \neq m$. It is exact if, and only if, C_m is also 0.

Remark III.5.

By abuse of notations, if the context if clear, we will denote all the applications between cochains d^n.

Definition III.11.

Let $C^\bullet$ and $D^\bullet$ two complexes of cochains.
A morphism $f^\bullet : C^\bullet \to C'^\bullet$ is a sequence of morphism $(f^n : C^n \to C'^n)_{n\in\mathbb{Z}}$ such that for every $n \in \mathbb{Z}$, $f^n \circ d^{n-1} = d'^n \circ f^{n-1}$, *ie* the following diagram commutes :

$$\begin{tikzcd}
\dots \arrow[r] & C^{n-1} \arrow[r, "d^{n-1}"] \arrow[d, "f^{n-1}"] & C^n \arrow[r, "d^n"] \arrow[d, "f^n"] & C^{n+1} \arrow[r] \arrow[d, "f^{n+1}"] & \dots \\
\dots \arrow[r] & C'^{n-1} \arrow[r, "d'^{n-1}"] & C'^n \arrow[r, "d'^n"] & C'^{n+1} \arrow[r] & \dots .
\end{tikzcd}$$

With this construction, complexes of cochains (resp. bounded above, resp. bounded below, resp. bounded) form a category, denoted by $\mathrm{Comp}^\bullet(\mathscr{C})^\sharp$, with $\sharp = \emptyset, -, +, b$ respectively.

III.1.4.b. Cocycle, coboundary, cohomology

Definition III.12.

Let $C^\bullet$ be a complex of cochains.

1) The objects $\mathrm{Z}^n(C^\bullet) := \mathrm{Ker}(d^n)$ are called n-*cocycles* of the complex,
2) The objects $\mathrm{B}^n(C^\bullet) := \mathrm{Im}(d^{n-1})$ are called n-*coboundaries* of the complex,
1) The objects $\mathrm{H}^n(C^\bullet) := \mathrm{Z}^n(C^\bullet)/\mathrm{B}^n(C^\bullet)$ are called n-*cohomologies* of the complex.

Lemma III.4.

$\mathrm{Z}^n, \mathrm{B}^n$ and H^n induces covariant functors on $\mathrm{Comp}^\bullet(\mathscr{C})$, and there is an exact short sequence of functors :

$$0 \to \mathrm{B}^n \to \mathrm{Z}^n \to \mathrm{H}^n \to 0.$$

Definition III.13.

H^n is called the *n-th cohomology functor*.

Definition III.14.

A morphism $f : C^\bullet \to D^\bullet$ is called a *quasi-isomorphism* if it induces an isomorphism in cohomology : for every $n \in \mathbb{Z}$,

$$\mathrm{H}^n(f) : \mathrm{H}^n(C^\bullet) \xrightarrow{\sim} \mathrm{H}^n(D^\bullet).$$

The cohomology objects are a mean to measure the exactness default of a cochain at a certain degree : $\mathrm{H}^n(C^\bullet) = 0$ if, and only if $C^\bullet$ is exact at degree n.
The cohomology functor is not exact (meaning, it does not transform a short exact sequence in another one) in general, but gives rise to a *long exact sequence in cohomology* :

Theorem III.5 ([Ass22], Theorem 5.12).

Let $0 \to C'^\bullet \xrightarrow{f^\bullet} C^\bullet \xrightarrow{g^\bullet} C''^\bullet \to 0$ a short exact sequence of complexes of cochains. There exists an exact sequence :

$$\cdots \to \mathrm{H}^n(C'^\bullet) \xrightarrow{\mathrm{H}^n(f^\bullet)} \mathrm{H}^n(C^\bullet) \xrightarrow{\mathrm{H}^n(g^\bullet)} \mathrm{H}^n(C''^\bullet) \xrightarrow{\delta^n} \mathrm{H}^{n+1}(C'^\bullet) \xrightarrow{\mathrm{H}^{n+1}(f^\bullet)} \mathrm{H}^{n+1}(C'^\bullet) \to \dots$$

III.1.4.c. Homotopy

Two complexes of cochains can have the same n-th cohomology object (for all n) without being isomorphic, and two morphisms can induce the same in the cohomology without beoing equal. An important relation between morphisms having inducing the same in cohomology is the *homotopy* :

Definition III.15.

Let $f^\bullet, g^\bullet : C^\bullet \to C'^\bullet$ two morphisms of complexes of cochains.
$f^\bullet$ and $g^\bullet$ are called homotopic if there exists a sequence $(s^n : C^n \to C'^{n-1})_{n\in\mathbb{Z}}$ such that for all $n \in \mathbb{Z}$, $f^n - g^n = s^{n+1}d^n + d'^{n-1}s^n$:

$$\begin{tikzcd}
\dots \arrow[r] & C^{n-1} \arrow[r, "d^{n-1}"] \arrow[d, shift right, "g^{n-1}"'] \arrow[d, shift left, "f^{n-1}"] & C^n \arrow[r, "d^n"] \arrow[d, shift right, "g^n"'] \arrow[d, shift left, "f^n"] \arrow[dl, dashed, "s^n"'] & C^{n+1} \arrow[r] \arrow[d, shift right, "g^{n+1}"'] \arrow[d, shift left, "f^{n+1}"] \arrow[dl, dashed, "s^{n+1}"'] & \dots \\
\dots \arrow[r] & C'^{n-1} \arrow[r, "d'^{n-1}"'] & C'^n \arrow[r, "d'^n"'] & C'^{n+1} \arrow[r] & \dots
\end{tikzcd}$$

Lemma III.6 ([Ass22], Lemma 5.14).

The homotopic relation on $\mathrm{Hom}_{\mathrm{Comp}^\bullet(\mathscr{C})}(C^\bullet, C'^\bullet)$ is an equivalence relation. Moreover, if $f^\bullet$ and $g^\bullet$ are homotopic, then $\mathrm{H}^n(f^\bullet) = \mathrm{H}^n(g^\bullet)$ for every $n \in \mathbb{Z}$.

III.1.4.d. Double complexes and total complexes

See [[Ben22], III§5].
For the construction of the hypercohomology III.5.2.b, we will need a "resolution of the resolution" : a double complex.

Definition III.16.

A *double cochain complex* $X^{\bullet\bullet}$ is a diagram of the form

$$\begin{tikzcd}
 & {} & {} & {} & \\
0 \arrow[r] & X^{(0,2)} \arrow[u, "d^v_{0,2}"] \arrow[r, "d^h_{0,2}"] & X^{(1,2)} \arrow[u, "d^v_{1,2}"] \arrow[r, "d^h_{1,2}"] & X^{(2,2)} \arrow[u, "d^v_{2,2}"] \arrow[r, "d^h_{2,2}"] & {} \\
0 \arrow[r] & X^{(0,1)} \arrow[u, "d^v_{0,1}"] \arrow[r, "d^h_{0,1}"] & X^{(1,1)} \arrow[u, "d^v_{1,1}"] \arrow[r, "d^h_{1,1}"] & X^{(2,1)} \arrow[u, "d^v_{2,1}"] \arrow[r, "d^h_{2,1}"] & {} \\
0 \arrow[r] & X^{(0,0)} \arrow[u, "d^v_{0,0}"] \arrow[r, "d^h_{0,0}"] & X^{(1,0)} \arrow[u, "d^v_{1,0}"] \arrow[r, "d^h_{1,0}"] & X^{(2,0)} \arrow[u, "d^v_{2,0}"] \arrow[r, "d^h_{2,0}"] & {} \\
 & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] &
\end{tikzcd}$$

where, for every pair (i, j), we have

$$d^h_{i+1,j} \circ d^h_{i,j} = 0,\ d^v_{i,j+1} \circ d^v_{i,j} = 0,\ d^v_{i+1} \circ d^h_{i,j} + d^h_{i,j+1} \circ d^v_{i,j} = 0.$$

Definition III.17.

Let $X^{\bullet\bullet}$ be a double cochain complex.
The *total complex* attached to $X^{\bullet\bullet}$ is the complex $\mathrm{Tot}(X^{\bullet\bullet}) = X^\bullet$, with

- $X^n := \bigoplus_{i+j=n} X^{i,j}$,
- $d_n = d^h + d^v : X^n \to X^{n+1}$.

III.1.5 Derived categories

Our use of the derived categories will be superficial in this text : they will only appear in VI, especially since we will not give proofs.
We refer the curious reader to the presentation in [[HTT08], Appendix B], in which are being given references for more detailed construction : here is only a very brief description.

III.1.5.a. Derived categories

Starting an abelian category $\mathscr{C}$, we define its $\sharp$-homotopy category as its category of $\sharp$-complexes quotiented out by the homotopy equivalence (with again $\sharp = \emptyset, -, +, b$).
It is not an abelian category anymore, so the theory of *triangluated triangle* is to be developped. To get the $\sharp$-derived category $D^\sharp(\mathscr{C})$, we make invertible quasi-isomorphisms by a *localization process* (similar to the one in ring theory).
When we work with derived categories, we will think of there objects as chain of complexes of cochains with some relations.

III.1.5.b. Derived functors

What will be more useful are *derived functors*, essential in the cohomological theory. We are sketching here the construction, following [[Ben22], Chapter 3].
The goal is to approximate a right(or left)-exact functor by an exact functor. Some tools of exactness could be then used. They are also the natural functors to consider when working with derived categories.
In this subsubsection, $\mathscr{C}$ is an abelian category.

Definition III.18.

1) An object P of $\mathscr{C}$ is said to be *projective* if $\mathrm{Hom}_{\mathscr{C}}(P, -)$ is an exact functor.
$\mathscr{C}$ is said to *have enough projective objects* if for every $X \in \mathrm{Ob}(\mathscr{C})$, there exists a projective object P and an epimorphism $f : P \to X$.
2) An object I of $\mathscr{C}$ is said to be *injective* if $\mathrm{Hom}_{\mathscr{C}}(-, I)$ is an exact functor.
$\mathscr{C}$ is said to *have enough injective objects* if for every $X \in \mathrm{Ob}(\mathscr{C})$, there exists a injective object I and a monomorphism $g : X \to I$.

As these two notions are dual, and that the injective aspect proves more useful in the cohomological setup later on, the remainder will focus on the latter.

Definition III.19.

Let $X \in \mathrm{Ob}(\mathscr{C})$.
A *right resolution* of X is a sequence $I^\bullet : \dots I^0 \to I^1 \to \dots$ equipped with a morphism $e : X \to I^0$ such that

$$0 \to X \xrightarrow{e} I^0 \to I^1 \to \dots$$

is an exact sequence.
If all the I's are injective, the resolution is said to be *injective*.

By induction, we have the following result :

Proposition III.7.

Assume that $\mathscr{C}$ has enough injectives.
Then every object of $\mathscr{C}$ has an injective resolution.

Let $F : \mathscr{C} \to \mathscr{D}$ henceforth be a covariant left exact functor, with $\mathscr{D}$ abelian, and assume that $\mathscr{C}$ has enough injectives.
We construct the right exact functor associated to F and describe some of its properties. First, a lemma that is useful to the proofs (and helps grasping an intuition behind them) :

Lemma III.8.

Let X and Y objects of $\mathscr{C}$, let $I^\bullet$ an injective resolution of X, $J^\bullet$ a (arbitrary) right resolution of Y, and let $f : X \to Y$.
Then, there exists a morphism of chains $f^\bullet : I^\bullet \to J^\bullet$, unique up to homotopy, such that the diagram commutes

$$\begin{array}{ccccccccc}
0 & \longrightarrow & X & \longrightarrow & I^0 & \longrightarrow & I^1 & \longrightarrow & \dots \\
 & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f^0} & & \downarrow{\scriptstyle f^1} & & \\
0 & \longrightarrow & Y & \longrightarrow & J^0 & \longrightarrow & J^1 & \longrightarrow & \dots
\end{array}$$

We can apply the functor F to $I^\bullet$, but the sequence obtained $0 \to F(X) \to F(I^0) \to \dots$ is not exact in general since F is only exact on the left.

Proposition III.9.

Let $X \in \mathrm{Ob}(\mathscr{C})$ and $I^\bullet, J^\bullet$ two injective resolutions of X.
Then for every $n \in \mathbb{N}$, $\mathrm{H}^n(F(I^\bullet)) \simeq \mathrm{H}^n(F(J^\bullet))$.

Definition III.20.

For every $n \in \mathbb{N}$, we define

$$\mathrm{R}^n F(X) := \mathrm{H}^n(F(I^\bullet))$$

for any injective resolution $I^\bullet$ of X.

Here is a list of some properties of this object :

Proposition III.10.

Let X be an object of $\mathscr{C}$.

1) $\mathrm{R}^0 F(X) \simeq F(X)$,
2) $\mathrm{R}^n F(X)$ is well defined, up to isomorphisms,
3) If X is projective, for every $n \geq 1, \mathrm{R}^n F(X) = 0$,
4) Each morphisms $f : X \to Y$ induces a canonical $\mathrm{R}^n F(f) : \mathrm{R}^n F(X) \to \mathrm{R}^n F(Y)$ for every $n \geq 0$,
5) For every $n \geq 0, \mathrm{R}^n F(-) : \mathscr{C} \to \mathscr{D}$ is an additive functor and $\mathrm{R}^0 F \simeq F$.
6) Let $0 \to X' \to X \to X'' \to 0$ a short exact sequence. There exists a long exact sequence in the derived world :

$$0 \to \mathrm{R}^0 F(X') \to \mathrm{R}^0 F(X) \to \mathrm{R}^0 F(X'') \to \mathrm{R}^1 F(X') \to \dots.$$

Definition III.21.

The functors defined in 5) are called the *nth right derived functors* of F.

A injective resolution is usually hard to find. An other class of objects that can be used to compute the derived functors are acyclic objects :

Definition III.22.

An object J of $\mathscr{C}$ is said to be *acyclic* if for every $n \geq 1$, $\mathrm{R}^n F(J) = 0$, and a resolution is said to be acyclic if all its objects are acyclic.

Proposition III.11.

Let $J^\bullet$ an acyclic resolution of X.
Then for every $n \in \mathbb{N}$, $\mathrm{R}^n F(X) \simeq \mathrm{H}^n(F(J^\bullet))$.

Remark III.6.

Why are acyclic resolutions useful to compute derived functors if they are build satifying a condition on the said derived functors ?
The answer is that sheaf-theoric properties on some resolutions can give the acyclicity condition, as *flabby sheaves* [[Voi02], Definition 4.33] or *fine sheaves* [[Voi02], Definition 4.35].

Back to double complexes, one have the following :

Proposition III.12.

Assume that all the columns (respectively all raws) of $X^{\bullet\bullet}$ are exact.
Then, $\mathrm{Tot}(X^{\bullet\bullet})$ is acyclic.

Examples of such derived functors can be found in III.5.

III.2 Sheaf theory

This section will contain more examples because I gave a talkon the subject in the $\mathcal{D}$-seminar.
A classic reference is [[Har77], II§1].
In the following, X is a topological space and $\mathcal{T}_X$ will denote its topology category.

Definition III.23.

Let $\mathscr{C}$ be a category (often $\mathscr{C} = \mathscr{S}et, \mathscr{A}b, \mathscr{R}ings$...).
A *presheaf* of $\mathscr{C}$ is a contravariant functor $\mathcal{F} : \mathcal{T}_X \to \mathscr{C}$.
Explicitly, it is the data of :

1. for every open set U of X, an object $\mathcal{F}(U)$ of $\mathscr{C}$,
2. for every couple of open sets $V \subseteq U$ of X, a morphism $\mathrm{res}_{U,V} : \mathcal{F}(V) \to \mathcal{F}(U)$ verifying :
    a. for every open set U of X, then $\mathrm{res}_{U,U} = \mathrm{id}_U$,
    b. for every open sets $U \subseteq V \subseteq W$ of X, then $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

The objects $\mathcal{F}(U)$ are often called *sections of* $\mathcal{F}$ *over* U, and for avoiding confusion, will we denoted $\Gamma(U, \mathcal{F})$.

Remark III.7.

If the context is clear, we will denote $\mathrm{res}_{U,V}(f) = f|_U$ (the notation is justified in the next lines).

Example III.8.

1) The sheaf of real valued continuous functions $C^0(-, \mathbb{R})$ with classical restriction functions $\mathrm{res}(U, V)(f) = f|_U$.
2) If X has an additional structure, the above example can be further specified : if X is a

differentiable manifold, we can consider $C^\infty(-,\mathbb{R})$, if X is a complex manifold, $\mathscr{O}(-,\mathbb{C})$ ...

3) If C is an object of $\mathscr{C}$, the constant presheaf $\underline{C}$. We will be intersted in $\underline{\mathbb{Z}}$ or $\underline{\mathbb{C}}$.
4) The sheaf of real valued bounded functions $B(-,\mathbb{R})$.

Presheaves comes from a category point of view, so we need morphisms :

Definition III.24.

A *morphism of presheaves* (over the same space X) $\varphi : \mathcal{F} \to \mathcal{G}$ is a natural transformation. More explicitly, it is the data, for each $U \in \mathcal{T}_X$, of a morphism $\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$ of $\mathscr{C}$ such that the following diagram commutes for every $U \subseteq V$:

$$\begin{tikzcd}
\mathcal{F}(V) \arrow[r, "\varphi_V"] \arrow[d, "\mathrm{res}^{\mathcal{F}}_{U,V}"'] & \mathcal{G}(V) \arrow[d, "\mathrm{res}^{\mathcal{G}}_{U,V}"] \\
\mathcal{F}(U) \arrow[r, "\varphi_U"] & \mathcal{G}(U)
\end{tikzcd}$$

Definition III.25.

A *sheaf* $\mathcal{F}$ is a presheaf satisfying the extra conditions for any open cover $\{U_i, i \in I\}$ of an open set U of X:

3. for every $f, g \in \mathcal{F}(U)$ such that : $\forall i \in I,\ f|_{U_i} = g|_{U_i}$, then $f = g$.
4. for every $(f_i) \in \prod_{i\in I} \mathcal{F}(U_i)$ such that : $\forall (i,j) \in I^2,\ f_i|_{U_i \cap U_j} = f_j|_{U_i\cap U_j}$, then there exists $f \in \mathcal{F}(U)$ with $f|_{U_i} = f_i$ for any $i \in I$.

Remark III.9.

For $U = \emptyset$, it says that $\mathcal{F}(\emptyset)$ is the terminal object of $\mathscr{C}$ (if such one exists). Indeed, we can take the open covering of $\emptyset$ by a family indexed by $\emptyset$ itself, and we then have an arrow $\mathcal{F}(\emptyset) \to$

Example III.10.

- 1) and 2) are indeed sheaves.
- 3) is not, because of the previous remark.
- 4) is not either, with the conterexample $\mathbb{R} = \bigcup_{n\in\mathbb{N}} (-n, n)$ and $f : x \mapsto x$.
- If X is Hausdorff, x_* is a point of X, (S, s) a pointed set, the *skyscraper sheaf* is given by $\mathrm{sky}_{x_*}(S) : U \mapsto \begin{cases} S & \text{if } x_* \in U \\ s & \text{otherwise.} \end{cases}$

Definition III.26.

We will denote by PSh_X, respectively Sh_X, the category of presheaves, respectively the full subcategory of sheaves, over X (implicitly taking values in a certain category).

One can ask what looks like the behavior of the sheaf very locally, on a point. Since a singleton is in general not a point, we need the following definition.

Definition III.27.

Let $\mathcal{F}$ be a (pre)sheaf.
We define the *stalk* at $x \in X$ as :

$$\mathcal{F}_x := \{(U,s) |\ x \in U,\ U \in \mathcal{T}_X,\ s \in \mathcal{F}(U)\} \big/ \sim ,$$

where $(U,s) \sim (V,t) \Leftrightarrow (\exists W \subset U \cap V) : s|_W = t|_W$.

Remark III.11.

From a categorical point of view, it is also the direct limit : $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$, where the limit is taken over open set U containing x.

It turns out that this localization principle can be used to transform a presheaf into a sheaf in a natural way (in the sense of what follows).

Definition III.28.

Let $\mathcal{F}$ be a presheaf.
We define for every open set U

$$\mathcal{F}^\sharp(U) = \left\{ s : U \to \coprod_{x \in U} \mathcal{F}_x \ \middle|\ \left\{ \begin{array}{l} \forall x \in U,\ s(x) \in \mathcal{F}_x \\ \exists V \subset U, \exists t \in \mathcal{F}(V) \text{ such that } \forall y \in V,\ s(y) = t(y) \end{array} \right. \right\}.$$

$\mathcal{F}^\sharp$ is called the *sheafification of* $\mathcal{F}$.

Proposition III.13 ([Har77], II§1, Proposition 1.2).

$\mathcal{F}^\sharp$ is a sheaf.

Example III.12.

$\underline{\mathbb{R}}^\sharp$ is the sheaf of locally constant functions, more precisely the sheaf of continuous functions $C^0(-,\mathbb{R})$, where $\mathbb{R}$ is equipped with the discrete topology.

The proof in the Harsthone is based on a universal property, that can be reformulated this way :

Proposition III.14.

$((-)^\sharp, \text{Forget})$ is an adjoint pair of functors between PSh_X and Sh_X.

Example III.13.

Suppose that $\mathscr{C}$ admits kernels and cokernels (this assumption holds whenever kernels and cokernels are written).
For $\varphi : \mathcal{F} \to \mathcal{G}$, we define $\mathrm{Ker}(\varphi)(U) := \mathrm{Ker}(\varphi_U)$, $\mathrm{Im}(\varphi)(U) := \mathrm{Im}(\varphi_U)$ and $\mathrm{Coker}(\varphi)(U) := \mathrm{Coker}(\varphi_U)$.
The first one is always a sheaf, but the two others are not in general. In the following, $\mathrm{Im}(\varphi)$ or $\mathrm{Coker}(\varphi)$ will always denote the sheafification of these presheaves.

Proposition III.15.

A complex $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ is exact if, and only if the induced complex on stalks $0 \to \mathcal{F}_x \to \mathcal{G}_x \to \mathcal{H}_x \to 0$ is exact for every $x \in X$.

Proof

.

□

Remark III.14.

In general, it is not equivalent of the exactness of the induced sequence over every open set U. One could take as a counterexample the exponential sequence on $\mathbb{C}$:

$$\begin{array}{ccccccccc} 0 & \to & \underline{\mathbb{Z}} & \to & \mathscr{O}_{\mathbb{C}} & \to & \mathscr{O}_{\mathbb{C}}^* & \to & 0, \\ & & & & f & \mapsto & \exp(2\pi i f) & & \end{array}$$

that is not exact over $\mathbb{C}^*$, because the map $\mathscr{O}_{\mathbb{C}^*} \to \mathscr{O}_{\mathbb{C}^*}^*$ is not surjective : the map $z \mapsto z$ does not belong to the image since log is not well defined over $\mathbb{C}^*$.
This default in the exactness of these short sequences actually gives rise to the important theory of *sheaf cohomology*, see III.5.1.

Proposition III.16.

[[Sza12], Construction 2.7.8] Let $X = \cup_{i \in I} U_i$ an open covering of X. For every $i \in I$, let $\mathcal{F}_i$ a sheaf on U_i.
Assume that for every $i \neq j \in I$, there exists an isomorphism

$$\theta_{i,j} : \mathcal{F}_i|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{F}_j|_{U_i \cap U_j},$$

satisfying the condition $\theta_{j,k} \circ \theta_{i,j} = \theta_{i,k}$ on $U_i \cap U_j \cap U_k$ for every triplet of distinct elements of I.
Then, there exists a sheaf $\mathcal{F}$ on X such that for every $i \in I, \mathcal{F}|_{U_i} = \mathcal{F}_i$. Moreover, this sheaf is unique, up to unique isomorphism.

III.3 Complex geometry

A good reference for this subsecion is [Huy04].

III.3.1 First definitions

See [[Huy04], 2.1].

Definition III.29.

Let X be a differentiable manifold of dimension $2n$.
An *holomorphic atlas* is a set $\{(U_i, \varphi_i), i \in I\}$ where $X = \bigcup_{i \in I} U_i$ is an open cover and $\varphi : U_i \to \mathbb{C}^n$ are homeomorphisms on their image, such that for all $(i,j) \in I^2$, the *transition map* $\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$ is holomorphic.
Such a couple (U_i, φ_i) is called a *local coordinate system*, or just *coordinates*.

We say that two holomorpic atlases $\{(U_i,\varphi_i), i\in I\}$ and $\{(\tilde{U}_j,\tilde{\varphi}_j), j\in J\}$ are *equivalent* if for all $(i,j)\in I\times J, \tilde{\varphi}_j\circ\varphi_i^{-1}:\varphi_i(U_i\cap\tilde{U}_j)\to\tilde{\varphi}_j(U_i\cap\tilde{U}_j)$.

Definition III.30.

A *complex manifold of dimension* n is a real manifold X of dimension $2n$ endowed with an equivalence class of holomorphic atlases.

Remark III.15.

In the following, unless explicitly stated otherwise, the complex manifold will be endowed with an representant of the equivalence class, denoted by $\{(U_i,\varphi_i), i\in I\}$.

Definition III.31.

Let X be a complex manifold and $f:X\to\mathbb{C}$.
We say that f is *holomorphic* if for all local coordinates (U,φ), $f\circ\varphi:\varphi(U)\to\mathbb{C}$ is holomorphic.

Definition III.32.

Let X be a complex manifold.
We denote by $\mathscr{O}_X$ the sheaf of holomorphic function on X, *ie* $\mathscr{O}_X : U\mapsto\{f:U\to\mathbb{C}, f \text{ holomorphic}\}$.

Remark III.16.

$\mathscr{O}_{X,x}$ will denote the stalk of $\mathscr{O}_X$ at x. It coincides with the germs of holomorphic functions at x. *Via* local coordinates (U,φ), with $\varphi(x_*)=0$ for some $x_*\in U$, we have an isomorphism $\mathscr{O}_{X,x}\simeq\mathscr{O}_{\mathbb{C}^n,0}$. It is an integral domain, and we denote by $Q(\mathscr{O}_{X,x})$ the fraction field associated.

More generally, if the codomain is also a complex manifold, we have the following definition :

Definition III.33.

Let X and Y be complex manifolds and $f:X\to Y$ continuous.
We say that f is *holomorphic* if for all local coordinates (U,φ) on X and (U',φ') on Y, $\varphi'\circ f\circ\varphi:\varphi(U\cap f^{-1}(U'))\to\varphi'(U')$ is holomorphic.

Definition III.34.

Let X be a complex manifold.
A *meromorphic function* on X is a map $f:X\to\bigsqcup_{x\in X}Q(\mathscr{O}_{X,x})$ so that, for all $x_*\in X$, there exists an open neighborhood $U\subset X$ of x_* and $g,h\in\mathscr{O}_U$ verifying : $(\forall x\in U),\ f_x=\frac{g}{h}$.
The sheaf of meromorphic functions is denoted $\mathcal{K}_X$.

Remark III.17.

If $U\subset X$ is a connected open subset, then $\mathcal{K}_X(U)$ is a field, and if X itself is connected, $\mathcal{K}(X)=\mathcal{K}_X(X)$ is called the function field of X.

III.3.2 Vector bundle

See [[Huy04], 2.2].

Definition III.35.

Let X be a complex manifold and r a positive integer.
A *holomorphic vector bundle* or rank r on X is a complex manifold E equipped with a holomorphic map $\pi : E \to X$ such that :

1. for all $p \in X$, the *fiber* $E_p := \pi^{-1}(p)$ is a complex vector space of dimension r,
2. there exists an open covering $X = \bigcup_{i \in I} U_i$ and biholomorphic maps $\psi_i : \pi^{-1}(U_i) \to U_i \times \mathbb{C}^r$, called the *trivialization maps*, satisfying :
   a. the following diagram commutes :

$$\begin{tikzcd} \pi^{-1}(U_i) \arrow[rr, "\psi_i"] \arrow[dr, "\pi"'] & & U_i \times \mathbb{C}^r \arrow[dl, "\mathrm{pr}_1"] \\ & U_i & \end{tikzcd}$$

   b. The induced map $E_p \to \mathbb{C}^r$ for any $p \in U_i$ is $\mathbb{C}$-linear.

If $r = 1$, we say that (E, π) is a *line bundle*.

Remark III.18.

The trivialization maps induce linear isomorphisms $\psi_{i,j} = \psi_i \circ \psi_j^{-1} : U_i \cap U_j \to \mathrm{GL}_r(\mathbb{C})$, and the collection $\{\psi_{i,j}, i, j \in I\}$ is called a cocycle.
In particular, it verifies the conditions :

i. $\psi_{i,i} = \mathrm{id}_{U_i}$,
ii. $\psi_{j,k} \circ \psi_{i,j} = \psi_{i,k}$.

Proposition III.17.

Let X be a complex manifold and let r be a positive integer.
The categories $\mathrm{VectBun}^r(X)$ and $\mathrm{LocFree}_r(X)$ are equivalent.

PROOF

We define

$$\mathcal{F} : \begin{array}{ccl} \mathrm{VectBun}_r(X) & \longrightarrow & \mathrm{LocFree}_r(X) \\ E & \longmapsto & \mathcal{F}_E \end{array},$$

where $\mathcal{F}_E$ is the sheaf of sections of E. It is well defined by ??.
It is a functor : given a morphism of vector bundles $f : E_1 \to E_2$, we define for each open set U of X the function

$$\varphi_U : \begin{array}{ccl} \mathcal{F}_1(U) & \longrightarrow & \mathcal{F}_2(U) \\ s & \longmapsto & f \circ s \end{array}.$$

φ clearly defines a morphism of sheaves.

In order to construct a quasi-inverse, let $\mathcal{F} \in \mathcal{S}_X$ and $\psi : \mathcal{F}(U_i) \to {\mathscr{O}_{U_i}}^{\oplus r}$ local trivialisations on an open cover $\{U_i, i \in I\}$ of X.

□

III.3.3 Connections

See [[Huy04], 4.2].

Definition III.36.

Let $\mathcal{E}$ be a locally free $\mathscr{O}_X$-modules.
A *connection on* $\mathcal{E}$ is a $\mathbb{C}$-linear application $\nabla : \mathcal{E} \to \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^1_X$ satisfying the *Leibniz relation* : for any local sections s and local function f, then

$$\nabla(fs) = df \otimes s + f\nabla(s).$$

In particular, it induces, for each integer $k > 0$, a $\mathbb{C}$-linear map

$$\begin{array}{rccl} \nabla^{(k)} : & \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^k_X & \longrightarrow & \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^{k+1}_X \\ & s \otimes \omega & \longmapsto & (-1)^k \nabla s \wedge \omega + s \otimes d\omega \end{array}$$

A connection ∇ is said to be *flat* (or *integrable*) if the *curvature* $K := \nabla^{(1)} \circ \nabla$ is zero.

Remark III.19.

In dimension 1, we have $\Omega^2_X = 0$, so all connections are flat.

Example III.20.

1) d is a connection on $\mathscr{O}_X$ by identifying $\mathscr{O}_X \otimes_{\mathscr{O}_X} \Omega^1_X \simeq \Omega^1_X$. It induces a connection on $\mathscr{O}^r_X$ acting coordinate-wise.
2) For every $\omega \in \Omega^1_X$, the map

$$\begin{array}{rccl} d + \omega : & \mathscr{O}_X & \longrightarrow & \Omega^1_X \\ & f & \longmapsto & df + \omega f \end{array}$$

is a connection. More generally, if ω is a $r \times r$ matrix with coefficients in Ω^1_X, the map $d + \omega$ is a connection on $\mathscr{O}^r_X$.

The following result says that these are the only ones on $\mathscr{O}^r_X$, and then the only ones locally on $\mathcal{E}$.

Lemma III.18.

Every connection ∇ on $\mathscr{O}^r_X$ is written $d + \omega$ for some ω as above.

Proof

For every $f \in \mathscr{O}_X$ and $h \in \mathscr{O}^r_X$, we have by the Leibniz rule :

$$(\nabla - d)(fh) = f\nabla(h) + df \otimes h - fdh - df \otimes h = f(\nabla - d)h.$$

Thus, $\nabla - d : \mathscr{O}^r_X \to \Omega^1_X$ is $\mathscr{O}_X$-linear and it's the multiplication by a matrix ω.

□

Remark III.21.

Connections are a reformulation of (systems) of differential equations, locally on a vector bundle.
Let $\nabla = d + \omega$ a connection on $\mathscr{O}_X^r$ (it is of this form by the previous lemma). We choose local coordinates $z^1, \dots, z^n$ on X. Then we can write

$$\omega(z) = \sum_{j=1}^{n} A_i(z) dz^j.$$

For a f holomorphic on the chart, we then have :

$$\nabla f = \sum_{j=1}^{n} \left(\frac{\partial f}{\partial z^j} + A_j f \right) dz^j,$$

and $\nabla f = 0$ is equivalent to :

$$\begin{cases} \dfrac{\partial f}{\partial z^1}(z) = -A_1(z) f(z), \\ \qquad\qquad \vdots \\ \dfrac{\partial f}{\partial z^n}(z) = -A_n(z) f(z), \end{cases}$$

which is a system of *partial* differential equations.

Definition III.37.

We define the category $\mathrm{LocFree}^{\nabla}(X)$ with

- locally free $\mathscr{O}_X$-modules equipped with flat connections as objects,
- morphisms of $\mathscr{O}_X$-modules $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ making

$$\begin{tikzcd} \mathcal{E}_1 \arrow[r, "\nabla_1"] \arrow[d, "\varphi"] & \mathcal{E}_1 \otimes_{\mathscr{O}_X} \Omega_X^1 \arrow[d, "\varphi \otimes 1"] \\ \mathcal{E}_2 \arrow[r, "\nabla_2"] & \mathcal{E}_2 \otimes_{\mathscr{O}_X} \Omega_X^1 \end{tikzcd}$$

  commutes, as morphisms between $(\mathcal{E}_1, \nabla_1)$ and $(\mathcal{E}_2, \nabla_2)$.

Here are some algebraic construction one can obtain with connections :

Proposition III.19.

Let $(\mathcal{E}_1, \nabla_1)$ and $(\mathcal{E}_2, \nabla_2)$ two connections on X. The following are also connections :

1) $(\mathcal{E}_1^*, (- \otimes 1) \circ \nabla_1)$,
2) $(\mathcal{E}_1 \oplus \mathcal{E}_2, \nabla_1 \oplus \nabla_2)$,
3) $(\mathcal{E}_1 \otimes_{\mathscr{O}_X} \mathcal{E}_2, \nabla_1 \otimes 1 + 1 \otimes \nabla_2)$
4) $(\mathcal{H}om_{\mathscr{O}_X}(\mathcal{E}_1, \mathcal{E}_2), \phi \mapsto (\nabla_2 \circ \phi) - (\phi \otimes 1) \circ \nabla_1)$

III.4 Algebraic geometry

vector bundle = loc free coherent sheaf

III.5 Some cohomology theory

For this section, we follow [[Har77], Chapter 3], [[Voi02], §4.3] and [HM17].
The cohomology theories are an entire world by themselves, especially for concrete computations (Čech, crystalline, étale, Dwork ...). We present only what is superficially needed (the proves would requiere more cohomology theories).

III.5.1 Sheaf cohomology

We will consider sheaf of abelian groups in this subsubsection.
As we notice in III.14, the functor $\Gamma := \Gamma(X, -) : \mathrm{Sh}_X \to \mathscr{A}b$ is not exact, but only left exact, for a topological space X. We want to apply tools from the derived functors to this particular functor.
First, one has to check that the category of sheaves of abelian groups satisfy the general assumptions, meaning that they have enough injectives :

> **Lemma III.20 ([Har77], Corollary 2.3).**
>
> The category Sh_X of sheaves of abelian groups has enough injectives.

Therefore, we can define :

> **Definition III.38.**
>
> We denote by $\mathrm{H}^n(X, \mathcal{F}) := \mathrm{R}^n\Gamma(\mathcal{F})$ the nth cohomology group of the sheaf $\mathcal{F}$.

III.5.2 de Rham cohomology

The two cohomologies presented will be linked by a theorem of Grothendieck later, see VII.2.2.a.

III.5.2.a. Analytic de Rham

In this paragraph, X is a complex manifold of dimension n.

> **Definition III.39.**
>
> The *de Rham complex* associated to X is :
>
> $$\Omega_X^\bullet : \mathscr{O}_X \xrightarrow{d} \Omega_X^1 \xrightarrow{d} \dots \xrightarrow{d} \Omega_X^n.$$

> **Theorem III.21 (Poincaré's lemma, [Voi02], Proposition 4.18).**
>
> $\Omega_X^\bullet$ is a right resolution of the constant sheaf $\mathbb{C}_X$.

We denote by $\mathrm{H}^\bullet_{\mathrm{dR}}(X)$ the associated cohomology groups.

> **Proposition III.22 ([Voi02], Corollary 4.37).**

We have the following :

$$\mathrm{H}^\bullet(X, \mathbb{C}) \simeq \mathrm{H}^\bullet_{\mathrm{dR}}(X).$$

III.5.2.b. Algebraic de Rham

In this paragraph, X is a complex manifold of dimension n.
The definition of the de Rham complex still holds (with algebraic differentials), but it does not give an acyclic resolution for the Zariski topology. Therefore, we use an exact resolution of this complex, and use the result III.12.

Definition III.40.

The *algebraic de Rham cohomology* of X is defined as the hypercohomology

$$\mathrm{H}^\bullet_{\mathrm{dR}}(X) = \mathbb{H}^i(X, \Omega^\bullet_X),$$

meaning the cohomology of the total complex associated to an exact resolution of $\Omega^\bullet_X$.

III.5.3 Singular cohomology

This theory originally comes from algebraic topology, with the objectives of detecting "holes" in a topological space. We follow [[Ass22], IX§6] and [[HM17], I§2.2].

Definition III.41.

Let $n \geq 0$ be a integer.
The *topological n-simplex* is defined as

$$\Delta_n := \{(t_0, \dots, t_n) \in \mathbb{R}^{n+1}, \forall i, t_i \geq 0, \text{ and } \sum_{i=0}^{n+1} t_i = 1\}.$$

Definition III.42.

A *singular n-simplex* is a continuous map

$$\Delta_n \to X.$$

The group of singular n-chains is the free abelian group :

$$S_n(X) = \mathbb{Z}[f \text{ singular } n\text{-simplex}]$$

and the group of singular n-cochains as

$$S^n(X) = \mathrm{Hom}_{\mathbb{Z}}(S_n(X), \mathbb{Z}).$$

Definition III.43.

The n-th boundary map is defined as

$$\begin{array}{rccl} d_n : & S_n(X) & \longrightarrow & S_{n-1}(X) \\ & f & \longmapsto & \sum_{i=0}^n (-1)^i f|_{i_i=0} \end{array}.$$

and the induced n-th coboundary map

$$d^n : S^n(X) \to S^{n+1}(X).$$

Lemma III.23 ([HM17], Lemma 2.2.3).

$(S_n(X), d_n)$ and $(S^n(X), d^n)$ are complexes of abelian groups.

Definition III.44.

We define singular homology and cohomology with values in $\mathbb{Z}$ are defined as

$$\mathrm{H}_i^{\text{sing}}(X, \mathbb{Z}) = \mathrm{H}_i(S_\bullet(X)), \quad \mathrm{H}^i_{\text{sing}}(X, \mathbb{Z}) = \mathrm{H}^i(S^\bullet(X)).$$

As the de Rham cohomology, relation exists with sheaf cohomology :

Theorem III.24.

Assume that X is locally contractible.
Then,

$$\mathrm{H}^i_{\text{sing}}(X, \mathbb{Z}) = \mathrm{H}^i(X, \mathbb{Z}).$$

IV The analytic Riemann-Hilbert correspondence

IV.1 Preliminary : a correspondence in algebraic topology

The goal of this subsection is to motivate the role of the Riemann-Hilbert correspondence in a bigger scheme, modeled on the Galois theory for field extensions, and introduce some notions that will be reused in a particular case later. We follow the presentation of [[Sza12], Chapter 2][1]. No proofs will be given.

IV.1.1 General considerations on cover theory

Definition IV.1.

Let X be a topological space.
A *covering space* over X is a topological space Y and a continuous $p : Y \to X$ such that : for each $x \in X$, there exists an open neighbourhood V of x and a discrete space S such that $\pi^{-1}(V) \simeq S \times V$.

Definition IV.2.

Let $p : Y \to X$ and $p' : Y' \to X$ two covering spaces.
A *morphism of cover* is a continuous $\phi : Y \to Y'$ such that $p' \circ \phi = p$.

One way of building such a covering is extracted from certains left-action of groups.

Definition IV.3.

Let G be a group acting continuously from the left on a topological space Y.
This action is said to be *even* if each point $y \in Y$ has some open neigbourhood U such that for each $g \neq h \in G, gU \cap hU = \emptyset$.

Proposition IV.1 ([Sza12], Lemma 2.1.7).

If G acts evenly on Y, the projection $p : Y \to G\backslash Y$ is a covering.

What follows is often refers as the Galois theory of cover : we study how the information of the (group of) automorphisms of a covering allows its classification.
We denote by $\mathrm{Aut}(Y/X)$ the group of automorphism of a cover $Y \to X$ (respecting the cover structure).
In order for the theory to works well, we assume that the base space X is locally connected.

Proposition IV.2 ([Sza12], Proposition 2.2.3).

Let $p : Y \to X$ be a connected cover.
Then the action of $\mathrm{Aut}(Y/X)$ on Y is even.

Conversely :

[1] the textbook actually explains this big scheme, often called the Galois-Grothendieck correpondence.

Proposition IV.3.

Let G be acting evenly on a connected space Y.
Then $\mathrm{Aut}(Y/(G\backslash Y)) = G$.

Finally, we can state the so-called *Galois correspondence* for covers.

Definition IV.4.

A connected cover $p : Y \to X$ is said to be a *Galois cover* if $\mathrm{Aut}(Y/X)$ acts transitively on each fiber of p.

Theorem IV.4 ([Sza12], Theorem 2.2.10).

Let $: Y \to X$ a Galois cover. We set $G = \mathrm{Aut}(Y/X)$.

1) For each subgroup H of G, the projection p induces a natural map $\bar{p}_H : H\backslash Y \to X$ which turns $H\backslash Y$ into a cover of X.
2) Let $q : Z \to X$ be a cover fitting in the diagram :

$$\begin{tikzcd} Y \arrow[r, "f"] \arrow[dr, "p"'] & Z \arrow[d, "q"] \\ & X \end{tikzcd}$$

then f is a Galois cover and $Z \simeq \mathrm{Aut}(Y/Z)\backslash Y$.

3) The maps $H \mapsto H\backslash Y$ and $Z \mapsto \mathrm{Aut}(Y/Z)$ induces a bijection between subgroups of G and intermediate cover of p as above,
4) The intermediate cover $q : Z \to X$ is Galois if, and only if, H is a normal subgroup of G, in which case $\mathrm{Aut}(Z/X) \simeq G/H$.

IV.1.2 The monodromy action

The monodromy action if often introduced in the context of differential equations (and it will the case in our study later on). However, it fits into the general theory of covers.
It starts with a "lifting" lemma :

Lemma IV.5 ([Sza12], Lemma 2.3.2).

Let $p : Y \to X$ be a cover, $y \in Y$ and set $x = p(y)$.

1) Let $f : [0,1] \to X$ be a path with base point x : $f(0) = x$. There exists a unique path $\tilde{f} : [0,1] \to Y$ such that $\tilde{f}(0) = y$ and $p \circ \tilde{f} = f$,
2) Let $g : [0,1] \to X$ be another path with base point x, homotopic to f. Then $\tilde{f}(1) = \tilde{g}(1)$ (for the unique $\tilde{g}$ as define above).

Therefore, we construct an action of the fundamental group $\pi_1(X, x)$ on the fiber $p^{-1}(x)$:

Definition IV.5 ([Sza12], Construction 2.3.3).

The *monodromy action* of $\pi_1(X,x)$ on $p^{-1}(x)$ as follows : if $\alpha = [f] \in \pi_1(X,x)$ and $y \in p^{-1}(x)$, then $\alpha.y := \tilde{f}(1)$.

Remark IV.1.

It's either a left or a right action, depending on the convention one adopt for the multiplication for paths. Let us assume that this is a left one.

This construction induces a functor :

Definition IV.6.

Let $x \in X$.
We define a functor

$$\begin{array}{rccl} \mathrm{Fib}_x : & \{p : Y \to X \text{ cover}\} & \longrightarrow & \{\text{set with a left-}\pi_1(X,x) \text{ action}\} \\ & p : Y \to X & \longmapsto & p^{-1}(x) \end{array},$$

the morphisms are respected by the definition of morphisms of covers.

Theorem IV.6 ([Sza12], Theorem 2.3.4).

Assume that X is connected and locally simply connected, let $x \in X$.
Then the functor Fib_x induces an equivalence of categories. Moreover, connected covers correspond to sets with a transitive action, and Galois cover to coset spaces of normal subgroups.

The proof comes with the construction of "the" *universal cover* :

Theorem IV.7 ([Sza12], Theorem 2.3.5).

Assume that X is connected and locally simply connected, let $x \in X$.
Then the functor Fib_x is representable by a cover $\tilde{X}_x$, called the *universal cover above x*.

In the other hand, in line with III.17, we have :

Proposition IV.8 ([Sza12], Theorem 2.5.9).

The functor $Y \mapsto \mathcal{F}_Y$ from the category of covers over X and the one of locally constant sheaves, obtained by taking the sheaf of sections over Y, induces an equivalence of categories.

Mixing both results, we can restrict the functor Fib_x to the following :

Theorem IV.9 ([Sza12], Corollary 2.6.2).

Assume that X is connected and locally simply connected, let $x \in X$.
The functor Fib_x induces an equivalence between the category of complex local systems, *ie* locally constant sheaves of complex vector spaces, and the one of finite dimensional left representations of $\pi_1(X,x)$.

Such a representation is called the *monodromy representation* of the local system.
It coincides with the monodromy in the sense of differential equations. Indeed, let us consider the following differential equation on a domain $U \in \mathbb{C}$:

$$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_n y = 0,$$

with the a_i's holomorphic on U. By a fundamental theorem from Cauchy, the solutions on an open neighourhood of each point is a finite dimensional vector space. Thus, the sheaf of solutions is a local system. By the previous theorem, it is associated to a monodromy representation of the fundamental group $\pi_1(U)$.

IV.2 Holomorphic version on a complex manifold

Some ideas of this paragraph are taken from [[Voi02], §9.2], [[Sza12], §2.7] and [[Poo25], §7.7].

Theorem IV.10 (Riemann-Hilbert correspondence, complex manifold case).

Let X be a complex manifold.
The functor

$$\begin{aligned} F: \quad \mathrm{LocFree}^{\nabla}(X) &\longrightarrow \mathrm{LocSys}(X) \\ (\mathcal{E}, \nabla) &\longmapsto \mathcal{E}^{\nabla} := \mathrm{Ker}(\nabla) \end{aligned}$$

induces an equivalence of categories.
Its inverse is given by

$$\begin{aligned} G: \quad \mathrm{LocSys}(X) &\longrightarrow \mathrm{LocFree}^{\nabla}(X) \\ \mathcal{A} &\longmapsto (\mathscr{O}_X \otimes_{\mathbb{C}} \mathcal{A}, d \otimes 1) \end{aligned}$$

Proof

Step 1 : G is well defined, that is : if $\mathcal{A} \in \mathrm{LocSys}(X)$, then $(\mathcal{E}, \nabla) := (\mathcal{A} \otimes_{\mathbb{C}} \mathscr{O}_X, 1 \otimes d) \in \mathrm{LocFree}^{\nabla}(X)$.

First remark that $d \otimes 1$ is well defined by extension of $\mathscr{O}_X \to \Omega^1_X$ with the tensor product. We also use the identification

$$(\mathcal{A} \otimes_{\mathbb{C}} \mathscr{O}_X) \otimes_{\mathscr{O}_X} \Omega^1_X \simeq \mathcal{A} \otimes_{\mathbb{C}} (\mathscr{O}_X \otimes_{\mathscr{O}_X} \Omega^1_X) \simeq \mathcal{A} \otimes_{\mathbb{C}} \Omega^1_X$$

by commutativity of the tensor product.
For a small open set such that $\mathcal{A}|_U \simeq \mathbb{C}^r$, we have $(\mathcal{A} \otimes \mathscr{O}_X)|_U \simeq \mathcal{A}|_U \otimes \mathscr{O}_U \simeq \mathbb{C}^r \otimes \mathscr{O}_U \simeq \mathscr{O}_U^{\oplus r}$.
Finally, we have :

$$\nabla^{(1)} \circ \nabla(a \otimes f) = \nabla^{(1)}((a \otimes 1) \otimes df) = \nabla(a \otimes 1) \wedge df + a \otimes d(df) = (a \otimes d1) \wedge df = 0.$$

We again used that the image of $a \otimes f$ in $\mathcal{E} \otimes_{\mathscr{O}_X} \Omega^1_X$ is identified with $(a \otimes 1) \otimes df$.
Step 2 : G is a functor.
If $\phi : A \to B$ is a morphism in $\mathrm{LocSys}(X)$, we have an induced $G(\phi) = \phi \otimes 1 : A \otimes_{\mathbb{C}} \mathscr{O}_X \to B \otimes_{\mathbb{C}} \mathscr{O}_X$ that makes the following commutes :

$$\begin{tikzcd} A \otimes_{\mathbb{C}} \mathscr{O}_X \arrow[r, "\phi\otimes 1"] \arrow[d, "1\otimes d"'] & A \otimes_{\mathbb{C}} \mathscr{O}_X \arrow[d, "1\otimes d"] \\ A \otimes_{\mathbb{C}} \Omega^1_X \arrow[r, "\phi\otimes 1"] & B \otimes_{\mathbb{C}} \Omega^1_X \end{tikzcd}$$

Step 3 : F is well defined, that is : if $(\mathcal{E}, \nabla) \in \mathrm{LocFree}(X)$, then $\mathcal{E}^\nabla \in \mathrm{LocSys}(X)$.
This is the most difficult part. Let us sketch out the proof.
The result is equivalent to saying that *locally*, the solutions of our system of differential equations given by ∇ is a complex vector space, with dimension equal to the rank of the system. One can then think of the Cauchy-Lipschitz (or Picard-Lindelöf) theorem. It indeed works when X is a curve because we have locally an ordinary differential equation. If $n > 1$, this theorem cannot be applied.
There is an alternative for this type of system : the Frobenius theorem. One corollary of this theorem is the following lemma, that implies immedialety the result.

Lemma IV.11 ([Voi02], Lemma 9.12).

Let $x \in X$.
There exists an open set U containing x such that the evaluation $\mathrm{ev}_x : \mathrm{Ker}(\nabla|_U) \to \pi^{-1}(x)$ is an isomorphism (of sheaves of complex vector spaces).

Step 4 : F is a functor.
If $\phi : (\mathcal{E}_1, \nabla_1) \to (\mathcal{E}_2, \nabla_2)$ a morphism in $\mathrm{LocFree}^\nabla(X)$, then it induces a morphism of sheaves $\phi : \mathrm{Ker}(\nabla_1) \to \mathcal{E}_2$, and its image is a subset of $\mathrm{Ker}(\nabla_2)$ by the relation $\nabla_2 \circ \phi = (\phi \otimes 1) \circ \nabla_1$.
Step 5 : $GF \simeq 1_{\mathrm{LocFree}^\nabla(X)}$.
Let $(\mathcal{E}, \nabla) \in \mathrm{LocFree}^\nabla(X)$. Set

$$\alpha_{(\mathcal{E},\nabla)} : \begin{array}{cl} \mathrm{Ker}(\nabla) \otimes_{\mathbb{C}} \mathscr{O}_X & \longrightarrow \mathcal{E} \\ s \otimes f & \longmapsto fs \end{array}.$$

It is clearly well defined and for $s \otimes f \in \mathrm{Ker}(\nabla) \otimes_{\mathbb{C}} \mathscr{O}_X$, we have, using the Liebniz rule :

$$\nabla(\alpha_{(\mathcal{E},\nabla)}(s \otimes f)) = \nabla(fs) = s \otimes df = \alpha(s \otimes 1) \otimes df = (\alpha \otimes 1)((1 \otimes d)(s \otimes f)),$$

so $\alpha_{(\mathcal{E},\nabla)}$ define a morphism in the category $\mathrm{LocFree}^\nabla(X)$.
Better, it defines a natural transformation. Let $\phi : (\mathcal{E}_1, \nabla_1) \to (\mathcal{E}_2, \nabla_2)$ a morphism in $\mathrm{LocFree}^\nabla(X)$. Then,

$$\phi(\alpha_{(\mathcal{E}_1,\nabla_1)}(f \otimes s)) = \phi(fs)$$

and

$$\alpha_{(\mathcal{E}_2,\nabla_2)}(GF(\phi)) = \alpha_{(\mathcal{E}_2,\nabla_2)}(F(\phi)(s) \otimes f) = f\phi(s).$$

This is equal because ϕ is a morphism of $\mathscr{O}_X$-modules.
To show that this is an isomorphism, let U be a small open neighborhood of x, such that $\mathrm{Ker}(\nabla|_U) \simeq \mathbb{C}^r$. Let $(s_1, \dots, s_r)$ a basis of $\mathrm{Ker}(\nabla|_U)$. Then, $(s_1 \otimes 1, \dots, s_r \otimes 1)$ is a basis of $\mathrm{Ker}(\nabla|_U) \otimes_{\mathbb{C}} \mathscr{O}_U$.
Moreover by taking U small enough, $(s_1, \dots, s_r)$ is also a $\mathscr{O}_U$-basis of $\mathcal{E}(U) \simeq \mathscr{O}_U^{\oplus r}$. Indeed, the determinant of the matrix (in any basis of $\mathcal{E}_U$) with j-th column $(s_j^{(i)})^T$ is not zero at x, so does not vanish in a small neighborhood of x.
Therefore, the matrix of $\alpha_{(\mathcal{E},\nabla)}$ in the basis $(s_1, \dots, s_r)$ (on both sides) is the identity, and is invertible.
Thus, $\alpha_{(\mathcal{E},\nabla)}$ is invertible on each stalk, and it is an isomorphism of sheaves.

Step 6 : $FG \simeq 1_{\mathrm{LocSys}(X)}$.
We clearly have for a local system $\mathcal{A}$ on X, $(\mathcal{A} \otimes_{\mathbb{C}} \mathscr{O}_X)^{1\otimes d} \simeq \mathcal{A} \otimes_{\mathbb{C}} \mathbb{C} \simeq \mathcal{A}$, and it behaves well with morphisms, such that it defines an isomorphism of functors.

$\square$

IV.3 Meromorphic version on a complex manifold

For a Riemann surface (meaning, $\dim(X) = 1$), the previous result gives us that any finite dimensional representation of $\pi_1(X, x_*)$ can be obtained as the monodromy reprensentation of some linear system of holomorphic differential equations.
At the begenning of the twentieth century, Hilbert stated, in one of his "23 problems of the century" (precisely, the twenty-first) a more precise question : is it possible to obtain such a finite dimensional representation of the fundamental group as the monodromy group of a differential equation with "moderated-growth solutions" around poles, for say X equal to $\hat{\mathbb{C}} \setminus S$, with S a finite set of points. We develop this point of view in this section.

IV.3.1 In dimension 1

IV.3.1.a. Elements of Fuchs theory

We basically follow Haefliger's lecture in [Bor+87], and the theory exposed is usually refered to as the *Fuchs theory*. As in III.2, we add some more examples and proofs, as I gave a talk on the subject during the internship.

We focus out study around each point of S. Without loss of generality, the case around $z = 0$ is sufficient. We will denote D_ε the disk centered at 0 of radius ε and D_ε^* the associated punctured disk at $z = 0$.

If we want to translate the definition of a connection on D_ε, we would say that such a connection is a locally free sheaf of modules $\mathcal{E}$ over the sheaf of rings of meromorphic function defined on D_ε with only a pole at 0, equipped with a map $\nabla : \mathcal{E} \to \mathcal{E} \otimes_{\mathscr{O}_{D_\varepsilon}} \Omega^1_{D_\varepsilon}$ with linearity and Leibniz propreties.
We can assume that $\mathcal{E}$ is free. Moreover, it's more convenient to work with the field of germs of meromorphic functions around 0 denoted K.
Having localised our study, we then won't work with sheaf in this paragraph. This shift of persective leads us to the following definition.

Definition IV.7.

A (germ of) *meromorphic connexion* at 0 is a finite dimensional vector space M over K equipped with a $\mathbb{C}$-linear operator $\nabla : M \to M$ satisfying the Leibniz rule :

$$\forall (h, u) \in K \times M : \nabla(hu) = \frac{dh}{dz}u + h\nabla u.$$

With this definition, a morphism of connection is of linear morphism commuting with connections.

Remark IV.2.

The ring $D[z^{-1}]$ of germs of differential operators with meromorphic coefficients acts on a meromorphic connection (M, ∇) by $Pu = \sum_{j=1}^{N} a_j \nabla^j u$ if $P = \sum_{j=1}^{N} a_j \frac{d^j}{dz^j}$ and $u \in M$.

Remark IV.3.

As explained in III.21, given a basis of M, a connection map on M is equivalent to a system of differential equations : a meromorphic connection is given by $\nabla = d - Adz$, where A is a matrix with coefficient in K, and vice versa.
A morphism between systems of equation given by matrices A and B respectively is then a matrix $M : D_\varepsilon^* \to M_n(K)$ such that $\frac{dM}{dz} = BM - MA$, and we say that two systems of differential equations are *equivalent* if $M(z)$ is invertible for any $z \in D_\varepsilon$ (or equivalently, the associate connections are isomorphic).
In the rest of the paragraph, we will juggle between these two definitions.

Let us compute the monodromy associated to a differential system $y' = Ay$ on D_ε.
First, recall that the universal cover of D_ε^* given by the exponential is $\{t \in \mathbb{C}, |e^{2i\pi t}| < \varepsilon\}$. Multi-valued functions are thus defined on this universal cover. t and $\tilde{}$ will belong to the universal cover and z to D_ε.
Let $\tilde{S}(t) = (\tilde{U}_1, \dots, \tilde{U}_n)$ a fundamental system of multi-valued solutions. A is 1-periodic on the universal cover, so $t \mapsto \tilde{S}(t+1)$ is also a solution. This, there exists a constant invertible matrix C such that $\tilde{S}(t+1) = \tilde{S}(t)C$.

Definition IV.8.

The matrix C defined above is called the *monodromy matrix* of the system.

To get back in the punctured disk, let Γ a matrix such that $C = \exp(2i\pi\Gamma)$. Then, $\tilde{S}(t)\exp(-2i\pi t\Gamma)$ is 1-periodic on the universal cover, and one can well define $\Sigma(z) = \tilde{S}(\frac{1}{2i\pi}\log(z))e^{\log(z)\Gamma}$. Finally, we set $S(z) = \Sigma(z)e^{\log(z)\Gamma}$.

One cannot expect to have a correspondence between the category of meromorphic connections and the one of representations of $\pi_1(D_\varepsilon^*)$, as the following example shows :

Example IV.4.

Define the two connections $\nabla_1 = d$ and $\nabla_2 = d - \frac{dz}{z^2}$. They are not equivalent since $z \mapsto 0$ is not invertible, but they both have the trivial monodromy (the solutions of ∇_2 are of the form $Ce^{-1/z}$).

This phenomenon can be intuited as resulting of a *bad* behavior of the solutions at $z = 0$, meaning that they have there sort of an essential singularity.
Thus, we introduce a *nice* class of functions.

Definition IV.9.

Let f be a multivalued function around 0.
f has *moderated growth* if, for all sector $S^\eta_{(a,b)} := \{z \in D_\eta^* \mid \arg(z) \in (a,b)\}$ over which f is defined, there exist $k \in \mathbb{N}$ and $C > 0$ such that :

$$\forall z \in S^\eta_{(a,b)},\ |f(z)| \leq C|z^{-k}|.$$

Example IV.5.

1) If f is a single-valued function, f has moderate growth if, and only if f is meromorphic on D_ε (for ε small enough).

2) z^α for $\alpha \in \mathbb{C}$, $\log(z)^m$ for $m \in \mathbb{N}$, have moderate growth.

The following result if very powerful : it gives you information on the behavior of the solutions of a differential equation just by checking a condition on the coefficients of the differential operator with a very simple computation.

Theorem IV.12 (Fuchs, [Bor+87], Theorem 1.1.1).

Let $P = \sum_{k=0}^{n} a_k(z) \frac{d^{n-k}}{dz^{n-k}}$ a differential operator on D_ε with the a_k's meromorphic.
The followings are equivalent :

i) the solutions of $Pu = 0$ have moderate growth,

ii) for every $1 \leq i \leq n$, $\frac{a_i}{a_0}$ has a pole of order at most i at 0.

These systems are those we are interested in for an expected correspondence.

Definition IV.10.

A differential operator is said to be *regular* if it satifies the conditions of the previous theorem.

Example IV.6.

As expected, the example given in IV.4 is not regular.

Theorem IV.13 ([Bor+87], Theorem 1.3.1).

Let $(E) : y' = Ay$ a differential system on D_ε. The followings are equivalent :

i) (E) is equivalent to a regular system,

ii) (E) is equivalent to a system $y' = \frac{B(z)}{z} y$ for some matrix B holomorphic on D_ε,

iii) (E) is equivalent to a system $y' = \frac{\Gamma}{z} y$ for some constant matrix Γ.

With this theorem, the equivalent definition in terms of connections is the following :

Definition IV.11.

Let (M, ∇) be a meromorphic connection.
It is said to be an *regular connection* if there exists a basis $(e_1, \ldots, e_n)$ of M over K and holomorphic functions $(b_{i,j}(z))$ such that :

$$\forall i \in [\![1, n]\!], \forall z \in D_\varepsilon,\ z\nabla e_i = -\sum_{j=1}^{n} b_{i,j}(z) e_j.$$

The following theorem is what we were aiming for.

Theorem IV.14 ([Bor+87], Corollary 1.3.2).

Two regular systems on D_ε are equivalent if, and only if, their monodromy is conjugated.

Stated like that, it is not as precise as we would have wanted for an equivalence of categories like in IV.2. The next paragraph will make it more accurate, based on the theorem above.

First, a result on the good behavior of algebraic construction (see III.19) with regularity :

Proposition IV.15 ([HTT08], Proposition 5.1.11).

Let M and N be regular meromorphic connections.
Then $\mathrm{Hom}_K(M, N)$ and $M \otimes_K N$ are regular.

PROOF

We do the full computation on $\mathrm{Hom}_K(M, N)$, equipped with $\nabla : \phi \mapsto \nabla_N \circ \phi - \phi \circ \nabla_M$. The second one is similar.
We set the following notations : $\dim(M) = m, \dim(N) = n$, equipped respectively with basis $(e_1, \ldots, e_m)$ and $(e'_1, \ldots, e'_n)$, $\phi = \sum_{i,j} \phi_{i,j} \varepsilon_{i,j}$, with $\varepsilon_{i,j}(e_k) = \delta_{i,k} e'_j$, and $\nabla_M = \sum_{i,j} \frac{1}{z} b_{i,j} \varepsilon^M_{i,j}, \nabla_N = \sum_{i,j} \frac{1}{z} c_{i,j} \varepsilon^N_{i,j}$ (the $\varepsilon^M_{i,j}$ and $\varepsilon^N_{i,j}$ are defined analoguously, and the index does not range on the same interval, but the reader can write down the details if not convinced).
Therefore :

$$\begin{aligned}
\nabla(\phi) &= \left(\sum_{i,j} \frac{1}{z} c_{i,j} \varepsilon^N_{i,j} \circ \sum_{k,l} \phi_{k,l} \varepsilon_{k,l} \right) - \left(\sum_{i,j} \phi_{i,j} \varepsilon_{i,j} \circ \sum_{k,l} \frac{1}{z} b_{k,l} \varepsilon^M_{k,l} \right) \\
&= \sum_{i,j,k,l} \frac{1}{z} \left(c_{i,j} \phi_{k,l} \varepsilon^N_{i,j} \circ \varepsilon_{k,l} \right) - \sum_{i,j,k,l} \frac{1}{z} \left(\phi_{i,j} b_{k,l} \varepsilon_{i,j} \circ \varepsilon^M_{k,l} \right) \\
&= \sum_{i,j,k,l} \frac{1}{z} c_{i,j} \phi_{k,l} \delta_{i,l} \varepsilon_{k,j} - \sum_{i,j,k,l} \frac{1}{z} \phi_{i,j} b_{k,l} \delta_{i,l} \varepsilon_{k,j} \\
&= \sum_{j,k} \frac{1}{z} \left(\sum_i c_{i,j} \phi_{k,i} - \phi_{i,j} b_{k,i} \right) \varepsilon_{k,j}.
\end{aligned}$$

One can see that the connection obtained is of the desired form.

□

IV.3.1.b. The local lifting

This paragraph is partially inspired by [[Sza12], §2.7], and [[HTT08], Lemma 5.2.19] for the lifting of morphisms.
The precise Riemann-Hilbert correspondence will be stated next paragraph, based on what we already did in the holomorphic case. In this one, we try to construct a unique (regular) meromorphic connection on D_ε given a holomorphic connection on D^*_ε.
We return to the language of sheaves. Indeed, according to a general fact on non-compact Riemann surfaces, a locally free $\mathscr{O}_{D^*_\varepsilon}$-module on D_ε is actually free (for example, see [[For81], Theorem 30.4]).
We will denote by $\mathscr{O}_{D_\varepsilon}(*0)$ the sheaf of meromorphic functions on D_ε with a pole at $z = 0$.

Proposition IV.16.

Let $(\mathcal{E}, \nabla)$ a holomorphic connection on D^*_ε.
There exists regular meromorphic connection $(\bar{\mathcal{E}}, \bar{\nabla})$ on D_ε which agrees with $(\mathcal{E}, \nabla)$ on D_ε.
This connection is unique up to isomorphisms (of meromorphic connections).

Proof

We can assume $\mathcal{E} = \mathscr{O}_{D_\varepsilon}^r$ and $\nabla = d - Adz$. Set $S(z)$ a fundamental system of solutions. Let C the monodromy matrix associated to ∇, $S(z)$ a fundamental system of solutions, and let B a matrix such that $C = \exp(2i\pi B)$. We consider the connection $\bar{\nabla} = d - \frac{B}{z}dz$. A fundamental system of solutions is given by $z^B = e^{B\log(z)}$.
Let $U(z) = S(z)z^{-B}$. Then

$$U(ze^{2i\pi}) = S(ze^{2i\pi})e^{-B\log(ze^{2i\pi})} = S(z)Ce^{-2i\pi B}z^{-B} = U(z).$$

Here, by an abuse of notation, $ze^{2i\pi}$ denotes the composition by the simple loop going around 0 counter-clockwise.
Therefore, $U(z)$ is single-valued, and then holomorphic on D_ε^*. Its inverse is too, so it defines a isomorphism between $(\mathcal{E}, \nabla)$ and $(\mathscr{O}_{D_\varepsilon}(*0)^r, \bar{\nabla})$.

Uniqueness comes from IV.14.

□

To make a functorial statement, we need additionally an extension of morphisms, we will directly do it for the global framework.

IV.3.1.c. Global study via patching

We will state the "meromorphic" Riemann-Hilbert on $\mathbb{P}^1_{\mathbb{C}}$ minus a finite set of points. We firstly introduce some global notation compared with what we did the last two paragraphs, and then state and proof the result.
Some ideas comes from [Sza12].

In this paragraph, we set $X = \mathbb{P}^1_{\mathbb{C}}$, $D = \{s_1, \ldots, s_k\} \subset X$ and $Y = X \setminus D$.
We denote by $\mathscr{O}_X(*D)$ the sheaf of meromorphic functions on X with poles along D.

Definition IV.12.

Let $\bar{\mathcal{E}}$ a locally free $\mathscr{O}_X(*D)$-modules.
A *meromorphic connection* on $\bar{\mathcal{E}}$ is a $\mathbb{C}$-linear map $\bar{\nabla} : \bar{\mathcal{E}} \to \bar{\mathcal{E}} \otimes_{\mathscr{O}_X(*D)} \Omega^1_X$ satisfying the Leibniz rule.

Definition IV.13.

We define the category $\mathrm{LocFree}^{\bar{\nabla}}_{\mathrm{reg}}(X, D)$ with

- locally free $\mathscr{O}_X(*D)$-modules equipped with a regular meromorphic (flat) connection as objects,
- morphisms of $\mathscr{O}_X(*D)$-modules commuting with the connections as morphisms.

Theorem IV.17.

Let $(\mathcal{E}, \nabla)$ a holomorphic connection on Y.
There exists a regular meromorphic connection $(\bar{\mathcal{E}}, \bar{\nabla})$ on X which agrees with $(\mathcal{E}, \nabla)$ on Y.
This connection is unique up to isomorphisms (of meromorphic connections).

Proof

Let D_i small disks around each s_i that do not meet. On each D_i, let $(\bar{\mathcal{E}}_i, \bar{\nabla}_i)$ the unique (up to isomorphisms) lifting of $(\mathcal{E}_i, \nabla_i) := (\mathcal{E}, \nabla)|_{D_i}$ given by IV.16. On an open set U containing the complementary of $\bigcup_{i=1}^k D_i$ and which does not meet S, we set $\bar{\mathcal{E}} = \mathcal{E}$.
Then, we can glue back the sheaves on $X = \bigcup_{i=1}^k D_i \cup U$ by III.16 (the isomorphisms $\theta_{i,j}$ are just given by the ones coming from the construction of the $\bar{\mathcal{E}}_i$, and the triple intersection condition is empty), which gives us the sheaf $\bar{\mathcal{E}}$ on X that agrees with $\mathcal{E}$ on Y.
The uniqueness is given by the one on each D_i and the one on the patching result.

□

Now, we deal with morphisms :

Proposition IV.18 ([HTT08], Lemma 5.2.19).

Let $\phi : (\mathcal{E}_1, \nabla_1) \to (\mathcal{E}_2, \nabla_2)$ a morphism between holomorphic connections on Y.
Then, there exists a morphism of meromorphic connections $\bar{\phi} : (\bar{\mathcal{E}}_1, \bar{\nabla}_1) \to (\bar{\mathcal{E}}_2, \bar{\nabla}_2)$ on X that restricts to ϕ.

We need two intermediate result.

Lemma IV.19.

The restriction map given by IV.17 induces a bijection :

$$\Gamma(X, \mathrm{Ker}(\bar{\nabla})) \xrightarrow{\sim} \Gamma(Y, \mathrm{Ker}(\nabla)).$$

Proof

The injectivity is clear. For the surjectivity, let $s \in \Gamma(Y, \mathrm{Ker}(\nabla))$. $\bar{\nabla}$ is regular, so s, a solution of the differential equation induced is single-valued as being defined on the whole of Y, and therefore has moderate growth - meaning that it can be extended to a meromorphic section on D. This concludes.

□

This one follows the definition.

Lemma IV.20.

Let $(\bar{\mathcal{E}}_1, \bar{\nabla}_1)$ and $(\bar{\mathcal{E}}_2, \bar{\nabla}_2)$ two connections on D_ε.
Then,

$$\Gamma(X, \mathcal{H}om_{\mathcal{O}_X(*D)}(\bar{\mathcal{E}}_1, \bar{\mathcal{E}}_2)^{\nabla}) \simeq \mathrm{Hom}_{\mathrm{LocFree}^{\bar{\nabla}}_{\mathrm{reg}}(X,D)}((\bar{\mathcal{E}}_1, \bar{\nabla}_1), (\bar{\mathcal{E}}_2, \bar{\nabla}_2)),$$

where the ∇ on the left-hand side is the one associated to the sheaf $\mathcal{H}om$ (see III.19).

Proof (IV.18)

Set $\mathcal{E} = \mathcal{H}om_{\mathcal{O}_Y}(\mathcal{E}_1, \mathcal{E}_2)$ and $\bar{\mathcal{E}} = \mathcal{H}om_{\mathcal{O}_X(*S)}(\bar{\mathcal{E}}_1, \bar{\mathcal{E}}_2)$ with the associate ∇ and $\bar{\nabla}$. By IV.15, $\bar{\mathcal{E}}$ is regular, so we can apply the first lemma. The second one thus gives a bijection :

$$\mathrm{Hom}((\bar{\mathcal{E}}_1, \bar{\nabla}_1), (\bar{\mathcal{E}}_2, \bar{\nabla}_2)) \simeq \mathrm{Hom}((\mathcal{E}_1, \nabla_1), (\mathcal{E}_2, \nabla_2)),$$

the surjectivity of which ensures the result.

□

Theorem IV.21.

The restriction functor

$$\mathrm{LocFree}^{\bar{\nabla}}_{\mathrm{reg}}(X, D) \to \mathrm{LocFree}^{\nabla}(Y)$$

induces an equivalence of categories.

Proof

The pseudo-inverse functor is given by the construction above, and all the other verifications are clear.

□

With the work done the previous subsection, we have the correspondence as wished :

Corollary IV.22 (Riemann-Hilbert correspondence, analytic meromorphic case, dimension 1).

There is an equivalence of categories :

$$\mathrm{LocFree}^{\bar{\nabla}}_{\mathrm{reg}}(X, D) \cong \mathrm{LocSys}(Y).$$

IV.3.2 Sketch of a proof in higher dimension

The classical reference of this subsubsection is [Del70], and the presentation by Malgrange in [[Bor+87], Part IV] and restructured in [[HTT08], §5.2].

For the general case, we work with a pair (X, D), where X is a connected complex manifold of dimension n and D a divisor of X. We set $Y = X \setminus D$.
The main goal is to build a unique regular meromorphic lifting of connection on X given a holomorphic one on Y (see [[HTT08], Theorem 5.2.17]), called the *Deligne's canonical extension*. Notice that the central point in the work of Deligne is to redefine the regularity condition for higher dimension. Two approachs are possible : reduce to the 1-dimensional case by verifying the regularity on the pullback of the connection by some $D(0,1) \hookrightarrow X$, or to control the growth of the solutions on the associated PDE. See [[Del70], II§4, Theorem 4.1] for details.
The existence is somehow similar to what we have already done in the one-dimensional case, meaning that the can patch local lifting, and actually restrict to the case where $Y = (D^*)^r \times D^{n-r}$, where $D = D(0, \varepsilon)$, and we translate the problem in terms of monodromy representation to construct a suitable connection. This simplification for the form of Y however suits for D a *normal crossing* divisor : a divisor D is said to be of normal crossing if for every point $p \in D$, there exists local coordinates $(x_1, \dots, x_n)$ of a neighbourhood U of p in X such that $D \cap U = \{x_1 \dots x_m = 0\}$.
Fortunatly, one can always refer back to this case by the Hironaka's resolution of singularities (see [Hir64]) : there exists a map $f : X' \to X$, proper and surjective, such that $D' := f^{-1}D$ is a normal crossing divisor, and the restriction of $f : X' \setminus D' \to X \setminus D$ is an isomorphism. Therefore, the pullback of a connection M on X, g^*M, can be extended to a regular meromorphic connection on X', and can give back a regular connection on X by applying some functors (for the details, see [HTT08], Theorem 5.2.20).

The uniqueness requires more constraints, namely the log-poles of the connection around D, and the real-part of the eigenvalues associated to the connection being in a lenght-1 interval. But these constraints are actually useful to glue back local lifting, and have some control over the morphisms between connections.

The result we finally get is the following (we do not state the original one in [Del70] because it's already mixed with the algebraic case) :

Theorem IV.23 (Riemann-Hilbert correspondence, analytic meromorphic case, [HTT08], Corollary 5.2.21).

Let X be a complex manifold and D a divisor on X. Set $Y = X \setminus D$.
Then we have an equivalence :

$$\mathrm{LocFree}^{\bar{\nabla}}_{\mathrm{reg}}(X, D) \cong \mathrm{LocSys}(Y).$$

V The algebraic Riemann-Hilbert correspondence

This section is rather short. The main reason is that it is based on the previous study on analytic space : thanks to the *GAGA principle*, briefly explained in the first subsubsection, one can, provided a correct translation of the analytic notion of regularity into an algebraic one.
Again, the original paper is the on by Deligne [Del70], and we take some clarification in [[HTT08], 5.3] and [[Bor+87], IV.7].

V.1 Elements of GAGA

"GAGA" usually refers to an article by Serre [Ser56], "Géométrie algébrique et géométrie analytique". The starting point is the following : given a projective algebraic variety over $\mathbb{C}$, one could consider its algebraic structure (with regular functions) or its analytic structure (with holomorphic functions). Both of these studies gave in general similar results. So the following question arose : is there a systematic way to translate results in the algebraic setup into analytic ones, and vice versa ?
This paradigm is still active in the current research. Here, we only state the results that we need. For more, the reader could take a look at [Ser56] or [[GR71], Exposé XII] (in the language of schemes).

Proposition V.1.

There exists a functor $(-)^{\mathrm{an}}$ from the category of complex algebraic variety to the category of analytic space.
From a complex algebraic variety $(X, \mathscr{O}_X)$ (with the Zariski topology), the analytic space X^{an} consists of the same set of points, but equipped with the Euclidian topology, and the sheaf of regular functions seen as holomorphic functions $\mathscr{O}_X^{\mathrm{an}}$. From a morphism $f : X \to Y$ of algebraic variety, we associate $f^{\mathrm{an}} : X^{\mathrm{an}} \to Y^{\mathrm{an}}$.

Moreover, there also exists a functor from the category of algebraic coherent sheaves on X and analytic coherent sheaves on X^{an}, via the map $\mathcal{F}^{\mathrm{an}} = \mathcal{F} \otimes_{\mathscr{O}_X} \mathscr{O}_X^{\mathrm{an}}$.

Theorem V.2 ([Ser56], Theorem 2,3).

Let X be a projective algebraic variety.
The analytification on coherent sheaves is an equivalence of categories.

Theorem V.3 ([Ser56], Theorem 1).

For every coherent sheaf $\mathcal{F}$ and every $i \geq 0$, we have :

$$\mathrm{H}^i(X, \mathcal{F}) \simeq \mathrm{H}^i(X^{\mathrm{an}}, \mathcal{F}^{\mathrm{an}}).$$

We only stated the Riemann-Hilbert correspondence for complex manifold so we only work with the following restriction :

Proposition V.4.

The analytification functor on algebraic varieties restrict to a functor between smooth algebraic varieties and complex manifolds.

V.2 Meromorphic Riemann-Hilbert on a smooth algebraic variety

In this subsection, X will denote a complex smooth algebraic variety.

Let us establish the framework. Consider $j : X \hookrightarrow V$ an open embedding of X into a smooth algebraic variery V, such that $D = V \setminus X$ is a divisor. We study integrable connections $(\mathcal{E}, \nabla)$ on X, meaning a vector bundle on X as defined in III.4, with a $\mathbb{C}$-linear map $\nabla : \mathcal{E} \to \mathcal{E} \otimes \Omega^1_X$ satisfying the Leibniz rule. Meromorphic connections are defined the same way as before with the sheaf $\mathcal{O}_V(*D)$, and so are morphisms between connections.
A fundamental initial observation is the following :

Proposition V.5 ([HTT08], Lemma 5.3.1).

The category of flat connection on X is equivalent to the category of meromorphic flat connections with poles along D.

This means that contrary to the analytic case where Deligne had to define some conditions, the lifting of a connection on X is unique (up to isomorphism) in the algebraic setup : it is based on the equality $\mathcal{O}_V(*D) = j_*\mathcal{O}_X$. Therefore, we only consider connection on X.
One of the main points here is to find a suitable definition of regularity for an integrable connection $(\mathcal{E}, \nabla)$ on X.
As the GAGA theorems work mainly on projective varieties, the first idea is to embed X into a projective one before applying out tools and restricting to X. It can be done by the Hironaka's resolution of singularities ([Hir64]) : there exists an open embedding $j : X \to \bar{X}$ into a projective $\bar{X}$ such that $Y := \bar{X} \setminus X$ is a normal crossing divisor. We thus define the following :

Definition V.1.

$(\mathcal{E}, \nabla)$ is said to be *regular* if $((j_*\mathcal{E})^{\mathrm{an}}, \nabla^{\mathrm{an}})$ is a regular meromorphic connection with poles along Y^{an}, in the analytic sense.
We denote by $\mathrm{Conn}_{\mathrm{reg}}(X)$ the category of regular flat connections.

This definition depends *a priori* on the embedding $\bar{X}$ chosen, but one can show that it actually does not, see [[Bor+87], §7.1].
We can provide an equivalent formulation, not involving an embedding. It's this definition on which is based [HTT08] :

Proposition V.6 ([Bor+87], Proposition 7.1.1).

$(\mathcal{E}, \nabla)$ is regular if, and only if for any morphism $i_C : C \to X$ from an algebraic smooth curve C, the connection $i_C^*(\mathcal{E}, \nabla)$ is regular.

With some additional work, and the use of GAGA V.2 among other things, we obtain the following :

Theorem V.7 ([Bor+87], Theorem 7.2.1 or [HTT08], Theorem 5.3.8).

Let X be a smooth algebraic variety.
The analytification functor induces an equivalence of categories :

$$\mathrm{Conn}_{\mathrm{reg}}(X) \simeq \mathrm{LocFree}^{\nabla}(X^{\mathrm{an}})$$

And directly, by the analytic case IV.23 :

Corollary V.8.

Let X be a smooth algebraic variety. Then there is an equivalence :

$$\mathrm{Conn}_{\mathrm{reg}}(X) \simeq \mathrm{LocSys}(X^{\mathrm{an}}).$$

In particular, we obtain a totally algebraic description of the fundamental group of X^{an}.

VI A higher generalization : the $\mathcal{D}$-module theory

VI.1 General theory of $\mathcal{D}$-modules

VI.1.1 Introduction

VI.1.1.a. First definitions

VI.1.1.b. Relationship with PDEs

VI.2 Some interesting functors

VI.3 Holonomicity

VII Periods

for introduction : picard fuschs equation in a historical situation : from periods depending of a parameter, construct a differential equations attached

VII.1 Prerequisites of periods theory

We will not be able to provide an introduction as appetizing as the theory of periods would call for. A document of Javier Fresán [Fre22] (in French) gives a historical approach (starting with Newton), and some modern developments of the theory.
Here, we only give the necessary knowledge to read the rest of the report.
We use [Fre22] and [HM17].

VII.1.1 A first approach

Period numbers form a class of number, bigger than $\bar{\mathbb{Q}}$, that is supposed to contain *every reasonable number* that one could encounter. The designation originally comes from the period of an oval, or the period of a pendulum.
For the first definition, we need the following.

Definition VII.1.

Let a set $S \subset \mathbb{R}^n$.
S is said to be $\mathbb{Q}$-*semi-algebraic* if it can be expressed as an intersection, union or complement of sets of the form $\{(x_1, \ldots, x_n), P(x_1, \ldots, x_n) \geq 0\}$ with $P \in \mathbb{Q}[X_1, \ldots, X_n]$.

Definition VII.2.

A complex number Z is said to be a *period* if its real part and its imaginary part can be expressed as absolutely convergent integrals with respect to the Lebesgue measure μ

$$\int_S f(x_1, \ldots, x_n) d\mu(x_1, \ldots, x_n),$$

with S a $\mathbb{Q}$-semi-algebraic set and $f \in \mathbb{Q}(x_1, \ldots, x_n)$.
We denote by $\mathcal{P} \subset \mathbb{C}$ the set of all periods.

$\mathcal{P}$ is clearly countable, so almost-every complex number is not a period.
Algebraically :

Theorem VII.1 ([Fre22], Lemma 2.2).

$\mathcal{P}$ is a subring of $\mathbb{C}$.

Some examples are $\bar{\mathbb{Q}}$, π, (multi-)Zeta values on the integer $\zeta(n)$, $\log(q)$ for positive $q \in \mathbb{Q}$ ... Actually, no explicit number is known *not to be* a period.

Period numbers can have different integral representations, for example

$$\pi = \int_{x^2+y^2 \leq 1} dxdy = \int_{\mathbb{R}} \frac{dx}{x^2+1}.$$

We cannot proceed without stating of the most important conjecture in the field (not very precisely though, see [[Fre22], §3.1] for a more precise statement) :

Conjecture VII.2 (Kontsevich-Zagier, 2001, [Fre22], Conjecture 3.1).

Every linear relation between periods comes from these operations on (absolutely convergent) integrals :

1) Linearity
2) Change of variables
3) Stokes formula.

VII.1.2 When geometers brought cohomology

We will not really motivate the introduction of cohomological tools used for the study of periods (partly because I don't know it very well myself). Some good references for it seem to be [[Fre22], §3.3], [And16] (both in French).

VII.2 The Gauss-Manin connection

VII.2.1 Coming from topology

We essentially use [[Voi02], Chapter 9].

Let $\mathscr{X}$ and S be complex manifolds.

Definition VII.3.

Let $\pi : \mathscr{X} \to S$ be a holomorphic map.
We say that $\mathscr{X} \xrightarrow{\pi} S$ is a *family of complex manifold* if ϕ is a proper submersion.

We recall that proper means that the inverse image of a compact set is compact.
We will denote by $\mathscr{X}_s$ the fiber $\pi^{-1}(s)$ above $s \in S$.
The following result gives us the differentiable local behavior of such a family.

Theorem VII.3 (Ehresmann, [Voi02], Proposition 9.3).

Let $\mathscr{X} \xrightarrow{\pi} S$ be a family of complex manifolds, and assume that S is contractible, with a based point O.
Then, there exists a diffeomorphism $T : \mathscr{X} \xrightarrow{\sim} \mathscr{X}_O \times S$ such that $\mathrm{pr}_2 \circ T = \pi$.

Remark VII.1.

This theorem only gives information on the differentiable structures. Indeed, there exists conter-examples for the analytic structure, see [[Poo25], Warning 4.9]. It can however be improved in this case, see [[Voi02], Proposition 9.5].

Let $\mathscr{X} \xrightarrow{\pi} S$ be a family of complex manifold. By the so-called *proper base change theorem*, we obtain :

Proposition VII.4 ([Dim04], Proposition 2.3.26).

For every $s \in S$ and every $i \geq 0$, we have

$$(\mathrm{R}^i\pi_*\mathbb{C}_{\mathscr{X}})_s \simeq \mathrm{H}^i(\mathscr{X}_s, \mathbb{C}).$$

S is a complex manifold, so it is locally contractible : for every $s \in S$, there exists an open neighourhood U of s which is locally contractible. By Ehresmann's theorem, there exists an isomorphism $T : \pi^{-1}(U) \to \mathscr{X}_s \times U$, such that $\pi = \mathrm{pr}_2 \circ T$. Thus, we have

$$(\mathrm{R}^i\pi_*\mathbb{C}_{\mathscr{X}})|_U \simeq \mathrm{H}^i(\mathscr{X}_s \times U, \mathbb{C}) \simeq \mathrm{H}^i(\mathscr{X}_s, \mathbb{C}) \simeq (\mathrm{R}^i\pi_*\mathbb{C}_{\mathscr{X}})_s,$$

as U is contractible, and the sheaf $\mathrm{R}^i\pi_*\mathbb{C}_{\mathscr{X}}$ is locally constant.
Therefore, $\mathrm{R}^i\pi_*\mathbb{C}_{\mathscr{X}}$ is a local system. By applying the Riemann-Hilbert correspondence, we get :

Definition VII.4.

The natural connection $1 \otimes d$ on $\mathrm{R}^i\pi_*\mathbb{C}_{\mathscr{X}} \otimes_{\mathbb{C}} \mathscr{O}_S$ is called the *Gauss-Manin connection associated to the family* $\mathscr{X} \xrightarrow{\pi} S$ and is denoted by ∇_{GM}.

VII.2.2 Coming from algebraic constructions

In this subsubsection, $\mathscr{X} \xrightarrow{\pi} S$ is a family of complex varities, meaning that π is smooth proper and S is assumed to be smooth over $\mathbb{C}$.

VII.2.2.a. With the algebraic de Rham cohomology

This paragraph is an attempt to unpack ideas from [[Blo16], §1.3].

Consider the the sheaf relative 1-forms

$$\Omega^1_{\mathscr{X}/S} := \Omega^1_{\mathscr{X}}/\pi^*\Omega^1_S$$

and the associated j-forms

$$\Omega^j_{\mathscr{X}/S} := \bigwedge^j \Omega^1_{\mathscr{X}/S}.$$

They correspond to the j-forms over $\mathscr{X}$ "modulo the ones coming from S". These j-forms are equipped with natural maps $p_j : \Omega^j_{\mathscr{X}} \to \Omega^j_{\mathscr{X}/S}$.
From there, we can construct the *relative de Rham complex*

$$\Omega^*_{\mathscr{X}/S} : 0 \to \mathcal{O}_X \xrightarrow{d} \Omega^1_{\mathscr{X}/S} \xrightarrow{d} \dots \xrightarrow{d} \Omega^n_{\mathscr{X}/S},$$

where the derivations are given by $d^j_{\mathscr{X}/S} = p_{j+1} \circ d^j_{\mathscr{X}}$.
The analytification of this complex gives a resolution of the sheaf $\pi^{-1}\mathscr{O}_{S^{\mathrm{an}}}$ (we take the same notation π for π^{an}. Indeed, we have the following :

$$\mathrm{Ker}(p_1 \circ d^0_{\mathscr{X}^{\mathrm{an}}}) = (d^0_{\mathscr{X}^{\mathrm{an}}})^{-1}(\mathrm{Ker}(p_1)) = (d^0_{\mathscr{X}^{\mathrm{an}}})^{-1}(\pi^*\Omega^1_{S^{\mathrm{an}}}) = \pi^{-1}(d^0_{\mathscr{X}^{\mathrm{an}}})^{-1}\Omega^1_{S^{\mathrm{an}}} = \pi^{-1}\mathscr{O}_{S^{\mathrm{an}}}.$$

Therefore, we have the equality

$$\mathrm{R}^\bullet\pi_*(\Omega^\bullet_{\mathscr{X}^{\mathrm{an}}/S^{\mathrm{an}}}) \simeq \mathrm{R}^\bullet\pi_*(\pi^{-1}\mathscr{O}_{S^{\mathrm{an}}}).$$

By writing $\pi^{-1}\mathscr{O}_{S^{\mathrm{an}}} = \pi^{-1}\mathscr{O}_{S^{\mathrm{an}}} \otimes_{\pi^{-1}\mathbb{C}^{\mathrm{an}}_S} \mathbb{C}_{\mathscr{X}^{\mathrm{an}}}$ and using the projection formula, we get :

$$\mathrm{R}^\bullet\pi_*(\Omega^\bullet_{\mathscr{X}^{\mathrm{an}}/S^{\mathrm{an}}}) \simeq \mathrm{R}^\bullet\pi_*(\mathbb{C}_{\mathscr{X}^{\mathrm{an}}}) \otimes_{\mathbb{C}_{S^{\mathrm{an}}}} \mathscr{O}_{S^{\mathrm{an}}}.$$

The left-hand side correpond to the *relative de Rham cohomology*, and is equipped with an integrable connection given on the right-hand side by $1 \otimes d$.

Proposition VII.5.

The above connection is the *Gauss-Manin connection* associated to $\mathscr{X}^{\text{an}}/S^{\text{an}}$.

The connection coincides (or at least, there are isomorphic) with the first definition because of the Riemann-Hilbert correpondence and the fact that both associated local systems have the same fibers. This reformulation is rather useful because of a result by Grothendieck :

Theorem VII.6 ([Gro66], Theorem 1).

The relative algebraic and analytic de Rham cohomology groups are isomorphic. In formulas :

$$(\mathrm{R}^\bullet \pi_* \Omega^\bullet_{\mathscr{X}/S})^{\text{an}} \simeq \mathrm{R}^\bullet \pi_*^{\text{an}} \Omega^\bullet_{\mathscr{X}^{\text{an}}/S^{\text{an}}}.$$

To descent the GM connection to the algebraic world, we still need to check that it is regular (and use V.7). This is a theorem of Griffiths :

Theorem VII.7 (Griffiths, [Gri70], Theorem 4.3).

The Gauss-Manin connection is regular.

Therefore, we have :

Proposition VII.8.

The relative algebraic de Rham cohomology is equipped with an algebraic Gauss-Manin connection.

VII.2.2.b. Using filtrations

The construction of this paragraph is taken from [KO68], and some retrospective clarifications in [[Hul25], §2.3] and [[Blo16], §1.3].

Start with the exact sequence coming from the definition :

$$0 \to \pi^*\Omega^1_S \to \Omega^1_{\mathscr{X}} \to \Omega^1_{\mathscr{X}/S} \to 0.$$

It induces a filtration on $\Omega^\bullet_X$:

$$F^p\Omega^n_X := \mathrm{Im}\left(\begin{array}{llll} : & \pi^*\Omega^p_S \otimes_{\mathcal{O}} \Omega^{n-p}_X & \longrightarrow \Omega^n_X \\ & \eta \otimes \alpha & \longmapsto \pi * \eta \wedge \alpha \end{array} \right).$$

VII.2.2.c. In the $\mathcal{D}$-module theory

VII.3 The variational theory of periods and the Picard-Fuchs equations

The situation is now the following : consider $k \subset \mathbb{C}$ a field, and a family $\mathscr{X} \xrightarrow{\pi} S$ of algebraic varieties over k.
We have at our disposal the complexified $\mathscr{X}_{\mathbb{C}} \xrightarrow{\pi_{\mathbb{C}}} S_{\mathbb{C}}$ and the analytified $\mathscr{X}^{\text{an}} \xrightarrow{\pi^{\text{an}}} S^{\text{an}}$ over $\mathbb{C}$.
We study the variation of the period numbers on the fibers $\mathscr{X}_s$ along S, by the use the previously

constructed Gauss-Manin connection.
The references are again fragments of [Blo16].

Consider a family $(\sigma_s)_{s\in S}$ of Betti classes and $(\omega_s)_{s\in S}$ of relative de Rham classes.
We obtain a function

$$F(s) = \int_{\sigma_s} \omega_s.$$

We want to understand the behavior of the variation of F.
Its external derivative is related to the Gauss-Manin in the following way :

Proposition VII.9.

$$d_S F(s) = \int_{\sigma_s} \nabla_{GM}(\omega_s)$$

Proof

The comparison isomorphism in VII.2.2.a gives

$$\mathrm{R}^\bullet \pi_*(\Omega^\bullet_{\mathscr{X}^{\mathrm{an}}/S^{\mathrm{an}}}) \simeq \mathrm{R}^\bullet \pi_*(\mathbb{C}_{\mathscr{X}^{\mathrm{an}}}) \otimes_{\mathbb{C}_{S^{\mathrm{an}}}} \mathscr{O}_{S^{\mathrm{an}}}.$$

In a local basis, this isomorphism has (j,k) matrix coefficients equal to $\int_{\sigma_{j,s}} \omega_{k,s} =: \langle \sigma_{j,s}, \omega_{k,s}\rangle$ and the derivation acts by Leibniz as $d_S\langle \sigma_{j,s}, \omega_{k,s}\rangle = \langle \sigma_{j,s}, \nabla_{GM}(\omega_{k,s})\rangle$ because $\sigma_{j,s}$ is a flat section.

$\square$

For a first explicit approach, let us assume that S is a curve. Therefore, noticing that the $\mathrm{H}^i_{\mathrm{dR}}(\mathscr{X}_s)$ are finite dimensional of dimension n over $\mathbb{C}_s$, so there exists a linear relation between $(\omega_s, \dots, \nabla^n_{GM}(\omega_s))$, say :

$$\sum_{j=0}^{n} a_j(s) \nabla^j_{GM}(\omega_s) = 0.$$

In the other side of the above equality, we get :

$$\sum_{j=0}^{n} a_j F^{(j)}(s) = \sum_{j=0}^{n} a_j \int_{\sigma_s} \nabla^j_{GM}(\omega_s) = 0.$$

We then obtain a linear differential equation for F.

When S is not a curve, we use the same idea as III.21, to obtain a system of partial differential equations.
Now that we have developped the formalism of $\mathcal{D}$-modules, we can write it in a nicer way.

VIII Conclusion and follow-ups

VIII.0.0.a. Irregular $\mathcal{D}$-modules and exponential periods

VIII.0.0.b. p-adic Hodge theory

VIII.0.0.c. p-adic Riemann-Hilbert correspondence

VIII.0.0.d. Arithmetic $\mathcal{D}$-modules

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